

High-order time-marching schemes for incompressible flow in particle methods

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ABSTRACT

This paper presents novel high-order time-marching schemes for simulating incompressible flow in particle methods. The proposed schemes are based on a newly developed formulation that describes the time evolution of computational variables along particle trajectories, resulting in a new form of the pressure Poisson equation. This formulation enables the direct application of existing forward-advancing time integration schemes, such as explicit Runge–Kutta methods, to achieve high-order temporal accuracy. Furthermore, the proposed schemes are generalized for arbitrary particle movement, enabling the efficient incorporation of particle shifting without requiring additional particle movement or variable corrections. By applying Runge–Kutta methods, this study presents four single- or multistage schemes referred to as RK1–RK4, corresponding to the number of stages. The validity of the proposed schemes is rigorously evaluated through numerical investigations involving four test cases and three types of particle movement (Lagrangian, Eulerian, and quasi-Lagrangian). The results reveal that the proposed RK2, RK3, and RK4 schemes achieve second-, third-, and fourth-order temporal convergence, respectively, and exhibit substantially higher accuracy than conventional first-order schemes, leading to improved volume and energy conservation. In addition, the proposed schemes demonstrate high computational efficiency, indicating their practical value for the numerical analysis of incompressible flow.

1. Introduction

Particle methods have been widely employed in various engineering applications owing to their ability to handle complex flows involving moving boundaries and free surfaces. Prominent particle methods include smoothed particle hydrodynamics (SPH) [1] and the moving particle semi-implicit (MPS) method [2]. Despite their versatility, these methods face challenges related to computational accuracy. Recent advancements in spatial discretization techniques, such as the least squares MPS (LSMPS) method [3], stabilized LSMPS method [4], compact MPS method [5], SPH(2) [6], and least squares SPH (LSSPH) [7], have considerably improved the accuracy of particle methods. In conventional particle methods, spatial discretization errors are dominant and, consequently, the accuracy of time integration has a limited influence on simulation quality. However, with the improved spatial accuracy achieved by advanced spatial discretization techniques, temporal accuracy has become a critical factor for further improving simulation quality. Despite its importance, the detailed characteristics of existing time-marching schemes and methods for achieving higher-order temporal accuracy in particle methods have not been thoroughly investigated.

In the context of incompressible flow simulations using particle methods, a critical requirement for time-marching schemes is the enforcement of the incompressibility condition. Two primary approaches are employed to achieve this: (i) maintaining a constant

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particle number density and (ii) ensuring a divergence-free velocity field. The MPS method, a pioneering semi-implicit particle method for incompressible flow proposed by Koshizuka and Oka [2], employs the former approach. Conversely, the incompressible SPH (ISPH) method introduced by Cummins and Rudman [8] employs the latter approach. While the particle number density approach exhibits favorable particle distribution uniformity and volume conservation, it often suffers from low accuracy. In contrast, the divergence-free velocity field approach generally offers higher accuracy but can become unstable when the particle distributions are distorted. To address the stability problem associated with the divergence-free velocity field approach, Xu et al. [9] proposed a quasi-Lagrangian method that incorporates particle shifting into the ISPH method. In recent developments of particle methods, approaches combining the divergence-free velocity condition with particle shifting are frequently employed due to their ability to achieve both high accuracy and stability. This study focuses on time-marching schemes for incompressible flow that utilize the divergence-free condition.

Most semi-implicit time-marching schemes used in particle methods are based on the fractional-step method. However, numerous variations exist due to differences in the order of updating particle positions and velocities, as well as in the definitions of the provisional velocity and position. For example, the scheme proposed by Koshizuka and Oka [2] first computes the provisional velocity, then moves particles to provisional positions, and finally corrects both the velocity and position using the pressure gradient. In contrast, the scheme proposed by Cummins and Rudman [8] updates particle positions before computing the provisional velocity. In addition, Lee et al. [10] presented a simplification of the scheme proposed by Cummins and Rudman, in which the initial particle movement is omitted. More recently, Matsunaga and Koshizuka [4,11] introduced a novel scheme that explicitly includes the convection term in the definition of the provisional velocity, similar to the fractional-step method employed in Eulerian frameworks.

Time-marching schemes are critical for achieving not only accuracy but also computational efficiency and numerical stability. In terms of efficiency, the number of particle movements per time step is an important factor. While many time-marching schemes involve two or more particle movements per time step, certain schemes, such as those proposed by Lee et al. [10] and Matsunaga and Koshizuka [4,11], require only one, making them relatively efficient. However, it has not been verified whether these schemes can achieve comparable accuracy. Furthermore, particle shifting generally requires additional particle movement and variable corrections, thereby impacting efficiency. In contrast, the scheme proposed by Matsunaga and Koshizuka [4,11] incorporates particle shifting within a single particle movement. Nevertheless, concerns remain regarding potential numerical instabilities due to the explicit convection term.

Research on achieving higher-order (greater than first-order) temporal accuracy is relatively limited. The aforementioned methods are based on the fractional-step method, which is essentially equivalent to the first-order accurate forward Euler method. For higher-order time-marching schemes used in the ISPH method, Hosseini and Feng [12] applied a second-order backward difference formula, while Trask et al. [13] employed the Crank–Nicolson method. In the context of weakly compressible SPH, higher-order explicit schemes, such as the fourth-order Runge–Kutta method, have been employed in numerous studies [14–17]. However, to the best of the authors' knowledge, rigorous numerical validations demonstrating high-order temporal convergence have not been reported.

In Eulerian mesh-based methods, high-order time-marching schemes for incompressible flow have been extensively studied using various projection-based multistage algorithms, typically explicit Runge–Kutta methods [18–21]. These methods have been demonstrated to achieve high-order temporal accuracy through analytical and numerical validation. However, the application of these multistage algorithms to particle methods is nontrivial. Particular difficulties arise from the need to account for particle movement, including particle shifting.

This study proposes novel high-order time-marching schemes for incompressible flow in particle methods. The central principle is to formulate the time evolution of the computational variables along particle trajectories and to derive a new form of the pressure Poisson equation that explicitly accounts for particle movement. With this new formulation, existing forward-advancing time integration schemes, such as Runge–Kutta methods, can be directly applied, achieving high-order temporal accuracy. Furthermore, the proposed formulation is generalized for arbitrary particle movement, enabling the efficient incorporation of particle shifting without requiring additional particle movement or variable corrections. The validity of the proposed schemes is evaluated through rigorous numerical investigations on several test cases employing sufficiently accurate spatial discretization to quantitatively evaluate temporal error behavior and the resulting accuracy.

The remainder of this paper is organized as follows. Section 2 describes the governing equations, boundary treatments, and spatial discretizations employed throughout this study, while Section 3 summarizes existing time-marching schemes in particle methods. Section 4 explains the derivation of the proposed time-marching schemes, followed by a detailed description of their computational procedures. Section 5 presents numerical validations of the proposed method, including comparisons with existing time-marching schemes and the finite volume method (FVM). Concluding remarks are provided in Section 6.

2. Governing equations and spatial discretization

This section describes the governing equations, boundary conditions, and spatial discretization techniques employed throughout this study. In particular, to enable a rigorous numerical assessment of temporal accuracy, an advanced high-order spatial discretization method, namely, the LSMPS scheme [3], is adopted. Boundary conditions are strictly enforced by directly incorporating them into the spatial discretization, as proposed in recent studies [4,22]. Only a brief overview is provided here, with detailed descriptions of the spatial discretization techniques and the free-surface identification algorithm provided in the literature cited.

2.1. Governing equations

This study considers incompressible flows of Newtonian fluids with constant density and viscosity. Therefore, the governing equations are the continuity and Navier–Stokes equations, given by

$$\nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla P + \nu \nabla^2 \mathbf{u} + \mathbf{g}, \quad (2)$$

where t denotes time, $\mathbf{u}$ denotes velocity, P denotes pressure, ρ denotes fluid density, ν denotes kinematic viscosity, and $\mathbf{g}$ denotes external force.

2.2. Free-surface boundary treatment

To ensure both computational accuracy and numerical stability, it is necessary to rigorously impose boundary conditions at the free surface. This involves the following steps: (i) labeling free-surface particles, (ii) determining the normal vector to the free surface, and (iii) enforcing the free-surface boundary conditions. For steps (i) and (ii), this study employs the free-surface identification method used in the stabilized LSMPS method [4]. This method combines the particle number density, a bias vector, and geometric scanning [23] to accurately detect free-surface particles. The normal vector to the free surface is determined using a surface fitting technique, which fits a polynomial surface to the positions of the free-surface particles and then computes the normal vector from the fitted surface.

The boundary conditions at the free surface are derived from the stress balance equations:

$$-P\mathbf{n} + \mu(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top) \cdot \mathbf{n} = \mathbf{f}_{\text{st}}, \quad (3)$$

where μ denotes viscosity, $\mathbf{f}_{\text{st}}$ denotes surface tension, and $\mathbf{n}$ denotes the outward unit normal vector to the free surface. For completeness, the surface tension term is included in Eq. (3), although surface tension effects are not considered in this study. By projecting Eq. (3) onto the normal direction $\mathbf{n}$ and the tangent plane, the following equations are derived:

$$P = \mu \mathbf{n} \cdot (\nabla \mathbf{u} + (\nabla \mathbf{u})^\top) \cdot \mathbf{n} - \mathbf{n} \cdot \mathbf{f}_{\text{st}}, \quad (4)$$

$$\mathbf{T} \cdot (\nabla \mathbf{u} + (\nabla \mathbf{u})^\top) \cdot \mathbf{n} = 0, \quad (5)$$

where $\mathbf{T}$ is a $(d-1) \times d$ matrix whose rows are orthogonal tangent vectors to the free surface, where d denotes the number of spatial dimensions. These equations serve as the pressure and velocity boundary conditions, respectively. Specifically, Eq. (4) defines the pressure at the free surface and is directly applied as a Dirichlet boundary condition for the free-surface particles within the pressure Poisson equation. In contrast, Eq. (5) provides a Neumann-type boundary condition for the velocity at the free surface, which is incorporated into the spatial discretization of the velocity for particles located near the free surface using the diffuse derivative constraint and the Lagrange multiplier method, as described in Section 2.3.

2.3. Spatial discretization

To ensure accurate evaluation of temporal errors, spatial errors must be sufficiently minimized. Consequently, a high-order spatial discretization scheme is employed—specifically, the LSMPS method [3]. This method is a well-established, high-order meshfree discretization scheme based on polynomial approximation and the weighted least squares method, exhibiting consistency even for nonuniform particle distributions. To minimize spatial errors in this study, a second- or fourth-order approximation is selectively applied, depending on the test case. The LSMPS method includes type-A and type-B formulations, which are employed according to the term being discretized. While type-A is typically employed, type-B is applied exclusively to the discretization of the pressure Poisson equation, a strategy known to improve numerical stability [4]. The following sections briefly describe the formulations of type-A and type-B LSMPS schemes, as well as the methods for incorporating additional constraints, such as boundary conditions, into these formulations.

2.3.1. Type-A LSMPS scheme

A p th-order polynomial approximation of a function ϕ around a particle $\mathbf{r}_i$ can be defined as

$$\mathcal{P}^\phi(\mathbf{r}, \mathbf{r}_i) := \phi_i + \mathbf{p}(\mathbf{r} - \mathbf{r}_i)^\top \mathbf{c}_i^\phi = \phi(\mathbf{r}) + \mathcal{O}(\|\mathbf{r} - \mathbf{r}_i\|^{p+1}), \quad (6)$$

where $\mathbf{p}$ is the polynomial basis vector and $\mathbf{c}_i^\phi$ is an undetermined coefficient vector:

$$\mathbf{p}(\mathbf{x}) := \left\{ \left(\frac{\mathbf{x}}{r_s} \right)^\alpha \mid 1 \leq |\alpha| \leq p \right\}, \quad (7)$$

$$\mathbf{c}_i^\phi := \left\{ \frac{r_s^{|\alpha|}}{\alpha!} D_\alpha^\phi \phi(\mathbf{x}) \Big|_{\mathbf{x}=\mathbf{r}_i} \mid 1 \leq |\alpha| \leq p \right\}. \quad (8)$$

Here, r_s is a scaling parameter ($r_s > 0$), while α is a d -dimensional multi-index, where d denotes the number of spatial dimensions.

By minimizing the weighted sum of squared residuals, the coefficient vector $\mathbf{c}_i^\phi$, and thus the spatial derivatives of ϕ at $\mathbf{r}_i$, are determined as

$$\mathbf{c}_i^\phi = \mathbf{M}_i^{-1} \mathbf{b}_i^\phi, \quad (9)$$

where $\mathbf{M}_i$ is the moment matrix and $\mathbf{b}_i^\phi$ is the source vector, defined as follows:

$$\mathbf{M}_i := \sum_{j \neq i} w_{ij} \mathbf{p}(\mathbf{r}_j - \mathbf{r}_i) \mathbf{p}(\mathbf{r}_j - \mathbf{r}_i)^\top, \quad (10)$$

$$\mathbf{b}_i^\phi := \sum_{j \neq i} w_{ij} \mathbf{p}(\mathbf{r}_j - \mathbf{r}_i) (\phi_j - \phi_i). \quad (11)$$

Here, w_{ij} denotes the weight function, defined as

$$w_{ij} = w_{\text{spike}}(\|\mathbf{r}_j - \mathbf{r}_i\|, r_e) := \begin{cases} \left(1 - \frac{\|\mathbf{r}_j - \mathbf{r}_i\|}{r_e}\right)^2, & \|\mathbf{r}_j - \mathbf{r}_i\| < r_e, \\ 0, & \|\mathbf{r}_j - \mathbf{r}_i\| \geq r_e, \end{cases} \quad (12)$$

where r_e denotes the effective radius.

In this study, the effective radius and the scaling parameter are defined based on a characteristic particle spacing Δx : specifically, $r_e = 3.1\Delta x$ for $p = 2$ and $r_e = 4.1\Delta x$ for $p = 4$, with $r_s = r_e/2$ for both cases.

2.3.2. Type-B LSMPS scheme

A key distinction between type-A and type-B LSMPS schemes lies in the treatment of the zeroth-order derivative. Specifically, the type-B scheme treats the function value at the center point ($\mathbf{r}_i$) as an undetermined coefficient. Accordingly, the polynomial approximation for the type-B scheme is defined as follows:

$$\hat{\mathbf{p}}^\phi(\mathbf{r}, \mathbf{r}_i) := \hat{\mathbf{p}}(\mathbf{r} - \mathbf{r}_i)^\top \hat{\mathbf{c}}_i^\phi = \phi(\mathbf{r}) + \mathcal{O}(\|\mathbf{r} - \mathbf{r}_i\|^{p+1}), \quad (13)$$

$$\hat{\mathbf{p}}(\mathbf{x}) := \left\{ \left(\frac{\mathbf{x}}{r_s} \right)^\alpha \middle| 0 \leq |\alpha| \leq p \right\}, \quad (14)$$

$$\hat{\mathbf{c}}_i^\phi := \left\{ \frac{r_s^{|\alpha|}}{\alpha!} D_x^\alpha \phi(\mathbf{x}) \middle|_{\mathbf{x}=\mathbf{r}_i} \middle| 0 \leq |\alpha| \leq p \right\}. \quad (15)$$

The coefficient vector $\hat{\mathbf{c}}_i^\phi$ is determined analogously to the type-A scheme, yielding the following equations:

$$\hat{\mathbf{c}}_i^\phi = \hat{\mathbf{M}}_i^{-1} \hat{\mathbf{b}}_i^\phi, \quad (16)$$

$$\hat{\mathbf{M}}_i := \sum_j w_{ij} \hat{\mathbf{p}}(\mathbf{r}_j - \mathbf{r}_i) \hat{\mathbf{p}}(\mathbf{r}_j - \mathbf{r}_i)^\top, \quad (17)$$

$$\hat{\mathbf{b}}_i^\phi := \sum_j w_{ij} \hat{\mathbf{p}}(\mathbf{r}_j - \mathbf{r}_i) \phi_j. \quad (18)$$

2.3.3. Diffuse derivative constraint

The diffuse derivative constraint [24] is a technique for constraining linear partial differential equations at arbitrary points around the center ($\mathbf{r}_i$). This technique is employed to incorporate the free-surface velocity boundary condition, Eq. (5), into the spatial discretization as a weak constraint for particles near the free surface, thereby improving numerical stability. The following describes the application of the diffuse derivative constraint to the type-A LSMPS scheme. An analogous approach is applicable to the type-B scheme; however, a detailed description is omitted for brevity.

According to the diffuse derivative approximation [25], a spatial derivative of a function ϕ at a point $\mathbf{r}_j$ can be approximated based on the polynomial approximation locally defined around $\mathbf{r}_i$:

$$(D_x^\beta \mathbf{p}^\phi)(\mathbf{r}_j, \mathbf{r}_i) := ((D_x^\beta \mathbf{p})(\mathbf{r}_j - \mathbf{r}_i))^\top \mathbf{c}_i^\phi = D_x^\beta \phi(\mathbf{x}) \Big|_{\mathbf{x}=\mathbf{r}_j} + \mathcal{O}(\|\mathbf{r}_j - \mathbf{r}_i\|^{p+1-|\beta|}), \quad (19)$$

where

$$(D_x^\beta \mathbf{p})(\mathbf{y}) := \left\{ D_x^\beta \left(\frac{\mathbf{x}}{r_s} \right)^\alpha \middle|_{\mathbf{x}=\mathbf{y}} \middle| 1 \leq |\alpha| \leq p \right\}, \quad (20)$$

and β denotes a d -dimensional multi-index.

Given a particle $\mathbf{r}_i$ and a set of neighboring free-surface particles Λ_i^{free} , the free-surface velocity boundary condition, Eq. (5), imposed at $\mathbf{r}_j$ ($j \in \Lambda_i^{\text{free}}$) can be approximated, for instance in two dimensions, as

$$\mathbf{T}_j \cdot \begin{bmatrix} 2(\partial_x \mathcal{P}^u)(\mathbf{r}_j, \mathbf{r}_i) & (\partial_y \mathcal{P}^u)(\mathbf{r}_j, \mathbf{r}_i) + (\partial_x \mathcal{P}^v)(\mathbf{r}_j, \mathbf{r}_i) \\ (\partial_y \mathcal{P}^u)(\mathbf{r}_j, \mathbf{r}_i) + (\partial_x \mathcal{P}^v)(\mathbf{r}_j, \mathbf{r}_i) & 2(\partial_y \mathcal{P}^v)(\mathbf{r}_j, \mathbf{r}_i) \end{bmatrix} \cdot \mathbf{n}_j = 0, \quad (21)$$

where $\mathbf{n}_j = [n_x, n_y]^\top$ denotes the normal vector at $\mathbf{r}_j$, and $\{u, v\}$ represents the velocity components. Substituting Eq. (19) into Eq. (21) yields a linear equation for the coefficient vectors $\{c_i^u, c_i^v\}$:

$$A_{ij} \begin{bmatrix} c_i^u \\ c_i^v \end{bmatrix} = 0, \quad (22)$$

$$A_{ij} := r_s \mathbf{T}_j \cdot \begin{bmatrix} 2n_x(\partial_x \mathbf{p})_{ij}^\top + n_y(\partial_y \mathbf{p})_{ij}^\top & n_y(\partial_x \mathbf{p})_{ij}^\top \\ n_x(\partial_y \mathbf{p})_{ij}^\top & n_x(\partial_x \mathbf{p})_{ij}^\top + 2n_y(\partial_y \mathbf{p})_{ij}^\top \end{bmatrix}. \quad (23)$$

Here, the abbreviations $(\partial_x \mathbf{p})_{ij} = (\partial_x \mathbf{p})(\mathbf{r}_j - \mathbf{r}_i)$ and $(\partial_y \mathbf{p})_{ij} = (\partial_y \mathbf{p})(\mathbf{r}_j - \mathbf{r}_i)$ are used for readability. It should be noted that the leading r_s on the right-hand side is intended for nondimensionalization.

The derived linear equation, Eq. (22), is incorporated into the LSMPS scheme as follows:

$$\begin{bmatrix} c_i^u \\ c_i^v \end{bmatrix} = \left(\begin{bmatrix} M_i & O \\ O & M_i \end{bmatrix} + \sum_{j \in \Lambda_i^{\text{free}}} \epsilon w_{ij} A_{ij}^\top A_{ij} \right)^{-1} \begin{bmatrix} b_i^u \\ b_i^v \end{bmatrix}, \quad (24)$$

where ϵ is a constant parameter controlling the strength of the constraint. In this study, $\epsilon = 10^{-3}$ is used to weakly impose the free-surface boundary condition near the free surface for stabilization, consistent with a previous study [4].

2.3.4. Lagrange multiplier method

While the diffuse derivative constraint imposes partial differential equations in a least-squares sense, the Lagrange multiplier method imposes them strictly, ensuring that particles satisfy the partial differential equations. In this study, the Lagrange multiplier method is applied to strictly impose the free-surface velocity boundary condition for free-surface particles and the pressure Poisson equation for internal particles. The formulation for applying the Lagrange multiplier method to the LSMPS scheme is described as follows.

A particle $\mathbf{r}_i$ located on the free surface must satisfy the free-surface velocity boundary condition given by Eq. (5). Accordingly, its coefficient vectors $\{c_i^u, c_i^v\}$ must satisfy the following linear equation:

$$A_{ii} \begin{bmatrix} c_i^u \\ c_i^v \end{bmatrix} = 0, \quad (25)$$

where the definition of A_{ii} is consistent with Eq. (23). To impose this condition, the Lagrange multiplier method is applied to the LSMPS scheme as follows:

$$\begin{bmatrix} c_i^u \\ c_i^v \\ \lambda_i \end{bmatrix} = \begin{bmatrix} M_i & O & A_{ii}^\top \\ O & M_i & A_{ii} \\ A_{ii} & O & 0 \end{bmatrix}^{-1} \begin{bmatrix} b_i^u \\ b_i^v \\ 0 \end{bmatrix}, \quad (26)$$

where λ_i denotes a set of Lagrange multipliers.

In projection-based time-marching schemes for incompressible flow, the pressure Poisson equation is solved to determine the pressure field. Although its specific formulation can vary depending on the selected time-marching scheme, for constant fluid density, it can generally be expressed as

$$\nabla^2 P = q, \quad (27)$$

where q denotes the source term. Typically, the pressure Poisson equation is solved by discretizing the Laplacian of pressure, yielding a linear system that is subsequently solved. However, this study employs a different approach, instead directly enforcing Eq. (27) as a constraint using the Lagrange multiplier method. This approach is known to provide numerical stabilization, particularly near free surfaces [4]. For the type-B LSMPS scheme, the corresponding constraint condition for the coefficient vector $\hat{c}_i^P$, with appropriate nondimensionalization, is given by

$$\boldsymbol{\vartheta}^\top \hat{c}_i^P = r_s^2 q_i, \quad (28)$$

$$\boldsymbol{\vartheta} = r_s^2 \left((\partial_x^2 \hat{\mathbf{p}})(0) + (\partial_y^2 \hat{\mathbf{p}})(0) \right). \quad (29)$$

The constrained type-B LSMPS scheme for the pressure Poisson equation is then given by

$$\begin{bmatrix} \hat{c}_i^P \\ \lambda_i \end{bmatrix} = \begin{bmatrix} \hat{M}_i & \boldsymbol{\vartheta}^\top \\ \boldsymbol{\vartheta} & 0 \end{bmatrix}^{-1} \begin{bmatrix} \hat{\mathbf{b}}_i^P \\ r_s^2 q_i \end{bmatrix}. \quad (30)$$

3. Existing time-marching schemes

3.1. Fractional-step method in an Eulerian framework

The fractional-step method is a fundamental time-marching technique for incompressible flow and serves a basis for numerous existing time-marching schemes in particle methods. Within Eulerian frameworks (i.e., without particle movement), the fractional-step method is computed directly through the following three steps:

(i) Compute the provisional velocity $\mathbf{u}^*$:

$$\mathbf{u}^* = \mathbf{u}^k + \Delta t(-(\mathbf{u}^k \cdot \nabla)\mathbf{u}^k + \nu \nabla^2 \mathbf{u}^k + \mathbf{g}). \quad (31)$$

(ii) Solve the pressure Poisson equation to obtain the pressure P^k :

$$\nabla^2 P^k = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*. \quad (32)$$

(iii) Update the velocity:

$$\mathbf{u}^{k+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla P^k. \quad (33)$$

Here, superscripts k and $k+1$ denote time steps, while Δt denotes the time step interval ($\Delta t = t_{k+1} - t_k$). It can be demonstrated that the above scheme exhibits first-order accuracy in time by comparing it with the Taylor series expansion:

$$\mathbf{u}(\mathbf{x}, t + \Delta t) = \mathbf{u}(\mathbf{x}, t) + \Delta t \left. \frac{\partial}{\partial t} \mathbf{u} \right|_{(\mathbf{x}, t)} + \mathcal{O}(\Delta t^2). \quad (34)$$

3.2. MPS-type time marching

The time-marching scheme employed in the MPS method [2] was initially developed based on the constant particle number density approach. However, it can be easily adapted to the divergence-free velocity field approach, and extensions incorporating a particle shifting technique are commonly employed. The resulting time-marching scheme can be summarized as follows:

(i) Compute the provisional velocity $\mathbf{u}^*$:

$$\mathbf{u}^* = \mathbf{u}^k + \Delta t(\nu \nabla^2 \mathbf{u}^k + \mathbf{g}). \quad (35)$$

(ii) Move particles to provisional positions $\mathbf{r}^*$:

$$\mathbf{r}^* = \mathbf{r}^k + \Delta t \mathbf{u}^*. \quad (36)$$

(iii) Solve the pressure Poisson equation to obtain the pressure P^{k+1} :

$$\nabla^2 P^{k+1} = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*. \quad (37)$$

(iv) Update the velocity and position:

$$\mathbf{u}^{k+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla P^{k+1}, \quad (38)$$

$$\mathbf{r}^{k+1} = \mathbf{r}^* - \frac{\Delta t^2}{\rho} \nabla P^{k+1}. \quad (39)$$

(v) Additional step for particle shifting (optional):

$$\mathbf{u}^{k+1} \leftarrow \mathbf{u}^{k+1} + \sum_{n=1}^p \frac{1}{n!} (\delta \mathbf{r} \cdot \nabla)^n \mathbf{u}^{k+1}, \quad (40)$$

$$\mathbf{r}^{k+1} \leftarrow \mathbf{r}^{k+1} + \delta \mathbf{r}, \quad (41)$$

where $\delta \mathbf{r}$ is the position shifting vector and p denotes the order of the polynomial approximation used for the velocity correction associated with particle shifting.

The aforementioned MPS-type time-marching scheme is formally of first order for both velocity and position, provided that the position shift $\delta \mathbf{r}$ is appropriately determined to preserve the free-surface shape. While this scheme has been widely adopted in the MPS method, several limitations require attention. First, the position update in step (ii) carries the risk of particle overlap and wall penetration because it uses the provisional velocity $\mathbf{u}^*$, which does not satisfy the divergence-free condition. This requires consideration of the Courant number for the provisional velocity, which often involves a smaller time step. Second, the pressure P^{k+1} obtained in step (iii) is not fully consistent with the Navier–Stokes equations. It is more accurately interpreted as a scalar potential for enforcing the incompressibility condition rather than as a physical pressure. Finally, when particle shifting is incorporated, this scheme requires three particle position updates per time step, which is disadvantageous in terms of computational efficiency.

3.3. ISPH-type time marching

The ISPH method [8] is the most widely used particle method for incompressible flow and has been extended with a particle shifting technique by Xu et al. [9]. The overall time-marching procedure is similar to that of the MPS method, with the main difference lying in the treatment of particle movement.

(i) Move particles to provisional positions $\mathbf{r}^*$:

$$\mathbf{r}^* = \mathbf{r}^k + \Delta t \mathbf{u}^k. \quad (42)$$

(ii) Compute the provisional velocity $\mathbf{u}^*$:

$$\mathbf{u}^* = \mathbf{u}^k + \Delta t (\nu \nabla^2 \mathbf{u}^k + \mathbf{g}). \quad (43)$$

(iii) Solve the pressure Poisson equation to obtain the pressure P^{k+1} :

$$\nabla^2 P^{k+1} = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*. \quad (44)$$

(iv) Update the velocity and position:

$$\mathbf{u}^{k+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla P^{k+1}, \quad (45)$$

$$\mathbf{r}^{k+1} = \mathbf{r}^k + \frac{\Delta t}{2} (\mathbf{u}^k + \mathbf{u}^{k+1}). \quad (46)$$

(v) Additional step for particle shifting (optional): Eqs. (40) and (41).

The ISPH-type time-marching scheme is formally first-order accurate for velocity and second-order accurate for position. The improved accuracy for position is due to the use of the midpoint velocity in Eq. (46). Similar to the MPS-type scheme, the ISPH-type scheme also involves three particle position updates per time step when particle shifting is included. Lee et al. [10] simplified this scheme by omitting the initial particle movement in (i). However, this simplification substantially reduces accuracy because it ignores inertial forces, as described in Section 5.

3.4. Split-type time marching

The velocity update from $(\mathbf{r}, t)$ to $(\mathbf{r} + \Delta \mathbf{r}, t + \Delta t)$ can be expressed by an infinite series:

$$\mathbf{u}(\mathbf{r} + \Delta \mathbf{r}, t + \Delta t) = \mathbf{u}(\mathbf{r}, t) + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\Delta t \frac{\partial}{\partial t} + \Delta \mathbf{r} \cdot \nabla \right)^n \mathbf{u}. \quad (47)$$

For a sufficiently smooth velocity field $\mathbf{u}(\mathbf{x}, t)$, the operators $\Delta t \partial/\partial t$ and $\Delta \mathbf{r} \cdot \nabla$ are commutative, allowing the infinite series to be factored into consecutive exponential operators:

$$\mathbf{I} + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\Delta t \frac{\partial}{\partial t} + \Delta \mathbf{r} \cdot \nabla \right)^n = \exp \left(\Delta t \frac{\partial}{\partial t} + \Delta \mathbf{r} \cdot \nabla \right) = \exp(\Delta \mathbf{r} \cdot \nabla) \exp \left(\Delta t \frac{\partial}{\partial t} \right). \quad (48)$$

By truncating the infinite series of each operator, the velocity update in Eq. (47) can be approximated as

$$\mathbf{u}(\mathbf{r} + \Delta \mathbf{r}, t + \Delta t) = \sum_{n=0}^p \frac{1}{n!} (\Delta \mathbf{r} \cdot \nabla)^n \left(\mathbf{u}(\mathbf{r}, t) + \Delta t \frac{\partial}{\partial t} \mathbf{u} \Big|_{(\mathbf{r}, t)} + \mathcal{O}(\Delta t^2) \right) + \mathcal{O}(\|\Delta \mathbf{r}\|^{p+1}). \quad (49)$$

Based on this formulation, the fractional-step method can be extended to account for particle movement [4,11]:

(i) Compute the provisional velocity $\mathbf{u}^*$:

$$\mathbf{u}^* = \mathbf{u}^k + \Delta t (-\mathbf{u}^k \cdot \nabla) \mathbf{u}^k + \nu \nabla^2 \mathbf{u}^k + \mathbf{g}. \quad (50)$$

(ii) Solve the pressure Poisson equation to obtain the pressure P^k :

$$\nabla^2 P^k = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*. \quad (51)$$

(iii) Update the velocity:

$$\mathbf{u}^{**} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla P^k. \quad (52)$$

(iv) Update the velocity and position:

$$\mathbf{u}^{k+1} = \mathbf{u}^{**} + \sum_{n=1}^p \frac{1}{n!} (\Delta \mathbf{r} \cdot \nabla)^n \mathbf{u}^{**}, \quad (53)$$

$$\mathbf{r}^{k+1} = \mathbf{r}^k + \Delta \mathbf{r}. \quad (54)$$

In this scheme, the particle displacement $\Delta \mathbf{r}$ can be selected arbitrarily. Consequently, quasi-Lagrangian particle movement with particle shifting $\delta \mathbf{r}$ can be efficiently achieved by setting

$$\Delta \mathbf{r} = \frac{\Delta t}{2} (\mathbf{u}^k + \mathbf{u}^{**} + \Delta t (\mathbf{u}^k \cdot \nabla) \mathbf{u}^k) + \delta \mathbf{r}, \quad (55)$$

where the convection term is added to approximate the midpoint velocity and to achieve second-order accuracy for position.¹ This split-type time-marching scheme can be interpreted as a direct extension of the fractional-step method for moving computational

¹ By using Eq. (50) and the Navier–Stokes equations, Eq. (52) can be transformed into $\mathbf{u}^{**} = \mathbf{u}^k + \Delta t \partial \mathbf{u}^k / \partial t$. Then, inserting Eq. (55) into Eq. (54) yields $\mathbf{r}^{k+1} = \mathbf{r}^k + \Delta t \mathbf{u}^k + (\Delta t^2/2) (\partial \mathbf{u}^k / \partial t + (\mathbf{u}^k \cdot \nabla) \mathbf{u}^k) + \delta \mathbf{r}$. In the absence of particle shifting, this gives an approximation of the Lagrangian particle motion, with a local truncation error of $\mathcal{O}(\Delta t^3)$, resulting in second-order accuracy for position.

points. In contrast to the MPS-type and ISPH-type schemes, this scheme involves only a single particle position update per time step, even when particle shifting is incorporated. Furthermore, given that the velocity extrapolation in Eq. (53) is sufficiently accurate, the scheme achieves the same temporal accuracy as Eulerian methods. However, explicitly incorporating the convection term into the provisional velocity raises concerns regarding numerical instability. Another critical problem is that this scheme lacks Galilean invariance, which can lead to substantial errors in the computation of fluids with large translational velocities (e.g., droplets and jets) and result in unphysical oscillations at the free surface.

4. Proposed time-marching scheme

4.1. Time evolution of velocity along particle trajectory

Let $\mathbf{r}_i(t)$ denote the trajectory of a particle with label i at time t . It is assumed that the moving velocity of the particle can be arbitrarily defined; that is, an arbitrary Lagrangian–Eulerian (ALE) framework is adopted. For example, in three typical cases—Eulerian, Lagrangian, and quasi-Lagrangian particle movement—the moving velocity is defined as follows:

$$\dot{\mathbf{r}}_i(t) := \frac{d}{dt} \mathbf{r}_i(t) = \begin{cases} \mathbf{0} & \text{(Eulerian)} \\ \mathbf{u}_i(t) & \text{(Lagrangian)} \\ \mathbf{u}_i(t) + \delta \mathbf{u}_i(t) & \text{(Quasi-Lagrangian)} \end{cases}, \quad (56)$$

where $\mathbf{u}_i(t) := \mathbf{u}(\mathbf{r}_i(t), t)$ denotes the flow velocity at the particle position, while $\delta \mathbf{u}_i$ denotes the shifting velocity associated with particle shifting, defined as $\delta \mathbf{u}_i = \delta \mathbf{r}_i / \Delta t$.

The time derivative of the flow velocity along the particle trajectory can be obtained using the chain rule:

$$\begin{aligned} \frac{d}{dt} \mathbf{u}_i(t) &= \frac{d}{dt} \mathbf{u}(\mathbf{r}_i(t), t) \\ &= \left. \frac{\partial}{\partial t} \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)} + \left. \frac{d}{dt} \mathbf{r}_i(t) \cdot \nabla \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)} \\ &= \left. \frac{D}{Dt} \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)} + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \mathbf{u}_i(t), \end{aligned} \quad (57)$$

where D/Dt denotes the material derivative and $\nabla \mathbf{u}_i(t)$ is the velocity gradient at the particle position, defined as

$$\nabla \mathbf{u}_i(t) := \left. \nabla \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)}. \quad (58)$$

Substituting the Navier–Stokes equations into Eq. (57) yields

$$\frac{d}{dt} \mathbf{u}_i(t) = -\frac{1}{\rho} \nabla P_i(t) + \nu \nabla^2 \mathbf{u}_i(t) + \mathbf{g}_i(t) + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \mathbf{u}_i(t), \quad (59)$$

where

$$\nabla P_i(t) := \left. \nabla P(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)}, \quad \nabla^2 \mathbf{u}_i(t) := \left. \nabla^2 \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)}, \quad \mathbf{g}_i(t) := \mathbf{g}(\mathbf{r}_i(t), t). \quad (60)$$

Eq. (59) is consistent with the Navier–Stokes equations expressed in a moving reference frame, as commonly employed in the ALE finite element method.

4.2. Pressure Poisson equation

To derive the pressure Poisson equation, consider the time derivative of the velocity divergence along the particle trajectory:

$$\begin{aligned} \frac{d}{dt} \nabla \cdot \mathbf{u}_i(t) &= \left. \frac{\partial}{\partial t} \nabla \cdot \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)} + \left. \frac{d}{dt} \mathbf{r}_i(t) \cdot \nabla \nabla \cdot \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)} \\ &= \left. \frac{D}{Dt} \nabla \cdot \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)} + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \nabla \cdot \mathbf{u}_i(t) \\ &= \nabla \cdot \left. \frac{D}{Dt} \mathbf{u}(\mathbf{x}, t) \right|_{\mathbf{x}=\mathbf{r}_i(t)} - \nabla \mathbf{u}_i(t) : \nabla \mathbf{u}_i(t) + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \nabla \cdot \mathbf{u}_i(t), \end{aligned} \quad (61)$$

where the following identity is used:

$$\nabla \cdot \frac{D}{Dt} \mathbf{u} = \frac{D}{Dt} \nabla \cdot \mathbf{u} + \nabla \mathbf{u} : \nabla \mathbf{u}. \quad (62)$$

The operator ‘:’ denotes the double dot product defined as $\mathbf{A} : \mathbf{B} = A_{ij} B_{ji}$ for second-order tensors $\mathbf{A}$ and $\mathbf{B}$. Substituting the Navier–Stokes equations into Eq. (61) yields

$$\frac{d}{dt} \nabla \cdot \mathbf{u}_i(t) = -\frac{1}{\rho} \nabla^2 P_i(t) + \nabla \cdot (\nu \nabla^2 \mathbf{u}_i(t) + \mathbf{g}_i(t)) - \nabla \mathbf{u}_i(t) : \nabla \mathbf{u}_i(t) + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \nabla \cdot \mathbf{u}_i(t). \quad (63)$$

By relating Eq. (63) to the incompressibility condition, the pressure Poisson equation can be derived.

For example, the first-order forward approximation of the velocity divergence at the new time step is expressed as

$$\nabla \cdot \mathbf{u}_i(t + \Delta t) = \nabla \cdot \mathbf{u}_i(t) + \Delta t \frac{d}{dt} \nabla \cdot \mathbf{u}_i(t) + \mathcal{O}(\Delta t^2). \quad (64)$$

Enforcing the incompressibility condition $\nabla \cdot \mathbf{u}_i(t + \Delta t) = 0$ and substituting Eq. (63) into Eq. (64) leads to the pressure Poisson equation for first-order forward-advancing time marching:

$$\frac{1}{\rho} \nabla^2 P_i(t) = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}_i(t) + \nabla \cdot (\nu \nabla^2 \mathbf{u}_i(t) + \mathbf{g}_i(t)) - \nabla \mathbf{u}_i(t) : \nabla \mathbf{u}_i(t) + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \nabla \cdot \mathbf{u}_i(t). \quad (65)$$

Eq. (65) is a general form of the pressure Poisson equation that explicitly accounts for particle movement. For $\dot{\mathbf{r}}_i(t) = \mathbf{0}$, it reduces to that in the Eulerian framework (i.e., Eq. (32)) because Eq. (65) can be rewritten as

$$\frac{1}{\rho} \nabla^2 P_i(t) = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}_i(t) + \nabla \cdot (\nu \nabla^2 \mathbf{u}_i(t) + \mathbf{g}_i(t) - \mathbf{u}_i(t) \cdot \nabla \mathbf{u}_i(t)) + \dot{\mathbf{r}}_i(t) \cdot \nabla \nabla \cdot \mathbf{u}_i(t), \quad (66)$$

using the following identity:

$$\nabla \cdot (\mathbf{u} \cdot \nabla \mathbf{u}) = \nabla \mathbf{u} : \nabla \mathbf{u} + \mathbf{u} \cdot \nabla \nabla \cdot \mathbf{u}. \quad (67)$$

In contrast, this equation does not correspond to any existing pressure Poisson equations employed in particle methods, whether for Lagrangian or quasi-Lagrangian particle movement. Furthermore, in ALE-type mesh-based methods, pressure equations are often derived from the divergence of the Navier–Stokes equations, resulting in a different form:

$$\frac{1}{\rho} \nabla^2 P_i(t) = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}_i(t) + \nabla \cdot (\nu \nabla^2 \mathbf{u}_i(t) + \mathbf{g}_i(t) + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \mathbf{u}_i(t)). \quad (68)$$

Therefore, Eq. (65) represents a novel form of the pressure Poisson equation for moving computational points. The above formulation of the pressure Poisson equation facilitates the development of high-order time-marching schemes for incompressible flow.

4.3. Multistage time integration

The derived equations for the time derivatives can be summarized as follows:

$$\frac{d}{dt} \Psi_i(t) = \mathbf{F}(t, \Psi_i(t)), \quad (69)$$

$$\Psi_i(t) := \begin{bmatrix} \mathbf{r}_i(t) \\ \mathbf{u}_i(t) \\ \nabla \cdot \mathbf{u}_i(t) \end{bmatrix}, \quad (70)$$

$$\mathbf{F}(t, \Psi_i(t)) := \begin{bmatrix} \dot{\mathbf{r}}_i(t) \\ -\frac{1}{\rho} \nabla P_i(t) + \nu \nabla^2 \mathbf{u}_i(t) + \mathbf{g}_i(t) + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \mathbf{u}_i(t) \\ -\frac{1}{\rho} \nabla^2 P_i(t) + \nabla \cdot (\nu \nabla^2 \mathbf{u}_i(t) + \mathbf{g}_i(t)) - \nabla \mathbf{u}_i(t) : \nabla \mathbf{u}_i(t) + (\dot{\mathbf{r}}_i(t) - \mathbf{u}_i(t)) \cdot \nabla \nabla \cdot \mathbf{u}_i(t) \end{bmatrix}. \quad (71)$$

Applying the Runge–Kutta method to these equations enables high-order accurate time integration, resulting in multistage schemes of second-order and higher, as described in the subsequent section. Each stage involves one particle position update and one velocity update, with the pressure Poisson equation solved in each stage to obtain intermediate pressures.

To ensure high-order accuracy through multistage schemes, particular attention must be paid to the pressure Poisson equations solved in the intermediate stages. Let $d^{(s)}$ denote the time derivative of the velocity divergence at the s -th stage of an S -stage scheme ($s = 0, 1, 2, \dots, S$):

$$d^{(s)} = -\frac{1}{\rho} \nabla^2 P^{(s)} + \nabla \cdot (\nu \nabla^2 \mathbf{u}^{(s)} + \mathbf{g}^{(s)}) - \nabla \mathbf{u}^{(s)} : \nabla \mathbf{u}^{(s)} + (\dot{\mathbf{r}}^{(s)} - \mathbf{u}^{(s)}) \cdot \nabla \nabla \cdot \mathbf{u}^{(s)}, \quad (72)$$

where $\mathbf{u}^{(s)}$ corresponds to $\mathbf{u}^{(0)} = \mathbf{u}^k$ and $\mathbf{u}^{(S)} = \mathbf{u}^{k+1}$ for the time marching from t_k to t_{k+1} . Time integration of the velocity divergence using a q -th order accurate forward-advancing scheme can be expressed as

$$\nabla \cdot \mathbf{u}^{k+1} = \nabla \cdot \mathbf{u}^k + \Delta t \sum_{s=0}^{S-1} \beta_s d^{(s)} + \mathcal{O}(\Delta t^{q+1}), \quad (73)$$

where β_s denotes the coefficients defined by the scheme. To ensure that the velocity divergence at the new time step satisfies the incompressibility condition (i.e., $\nabla \cdot \mathbf{u}^{k+1} = \mathcal{O}(\Delta t^{q+1})$), the following condition is enforced at each stage:

$$d^{(s)} = -\frac{1}{\Delta t} \nabla \cdot \mathbf{u}^k. \quad (74)$$

This leads to a new form of the pressure Poisson equation for multistage schemes:

$$\frac{1}{\rho} \nabla^2 P^{(s)} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^k + \nabla \cdot (\nu \nabla^2 \mathbf{u}^{(s)} + \mathbf{g}^{(s)}) - \nabla \mathbf{u}^{(s)} : \nabla \mathbf{u}^{(s)} + (\dot{\mathbf{r}}^{(s)} - \mathbf{u}^{(s)}) \cdot \nabla \nabla \cdot \mathbf{u}^{(s)}. \quad (75)$$

This equation is employed in the proposed time-marching schemes described below. Using this approach, the incompressibility condition can be satisfied with an error of $\mathcal{O}(\Delta t^{q+1})$, as demonstrated in Section 5.

Table 1

Coefficients for explicit Runge–Kutta methods. RK1 represents the forward Euler method, RK2 represents the modified Euler method, RK3 represents Heun's third-order method, and RK4 represents the classical fourth-order Runge–Kutta method.

				RK1	RK2	RK3	RK4
$\beta_{1,0}$				1	1/2	1/3	1/2
$\beta_{2,0}$	$\beta_{2,1}$				0	0	0
$\beta_{3,0}$	$\beta_{3,1}$	$\beta_{3,2}$				2/3	1/2
$\beta_{4,0}$	$\beta_{4,1}$	$\beta_{4,2}$	$\beta_{4,3}$			1/4	0
						3/4	1
							1/6
							1/3
							1/3
							1/6

4.4. Proposed scheme

The proposed time-marching schemes are constructed using explicit Runge–Kutta methods, and can be expressed as follows:

$$\Psi^{(0)} = \Psi^k, \quad (76)$$

$$\Psi^{(s)} = \Psi^k + \Delta t \sum_{n=0}^{s-1} \beta_{s,n} F(t^{(n)}, \Psi^{(n)}) \quad (s = 1, 2, \dots, S), \quad (77)$$

$$\Psi^{k+1} = \Psi^{(S)}, \quad (78)$$

where S denotes the number of stages, while $\beta_{s,n}$ denotes the coefficients defined by the Butcher tableau [26,27]. This study considers four variations of the Runge–Kutta method, as listed in Table 1, with the schemes labeled as RK1, RK2, RK3, or RK4 corresponding to the number of stages $S = \{1, 2, 3, 4\}$. It should be noted that the RK1 scheme corresponds to the first-order forward Euler method, which is not the primary focus of this study but is included for completeness and comparison purposes.

Using the above equations and the coefficients in Table 1, the proposed schemes compute time marching via the following procedure:

```

1:  $t^{(0)} \leftarrow t_k$ 
2:  $\mathbf{r}^{(0)} \leftarrow \mathbf{r}^k$ 
3:  $\mathbf{u}^{(0)} \leftarrow \mathbf{u}^k$ 
4: for  $s = 0, 1, 2, \dots, S - 1$  do
5:    $\dot{\mathbf{r}}^{(s)} \leftarrow \mathbf{u}^{(s)} + \delta \mathbf{u}$ 
6:    $\mathbf{h}^{(s)} \leftarrow \nu \nabla^2 \mathbf{u}^{(s)} + \mathbf{g}(\mathbf{r}^{(s)}, t^{(s)})$ 
7:   Solve  $\frac{1}{\rho} \nabla^2 P^{(s)} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^k + \nabla \cdot \mathbf{h}^{(s)} - \nabla \mathbf{u}^{(s)} : \nabla \mathbf{u}^{(s)} + (\dot{\mathbf{r}}^{(s)} - \mathbf{u}^{(s)}) \cdot \nabla \nabla \cdot \mathbf{u}^{(s)}$ 
8:    $\dot{\mathbf{u}}^{(s)} \leftarrow -\frac{1}{\rho} \nabla P^{(s)} + \mathbf{h}^{(s)} + (\dot{\mathbf{r}}^{(s)} - \mathbf{u}^{(s)}) \cdot \nabla \mathbf{u}^{(s)}$ 
9:    $t^{(s+1)} \leftarrow t^k + \Delta t \sum_{n=0}^s \beta_{s+1,n}$ 
10:   $\mathbf{r}^{(s+1)} \leftarrow \mathbf{r}^k + \Delta t \sum_{n=0}^s \beta_{s+1,n} \dot{\mathbf{r}}^{(n)}$ 
11:   $\mathbf{u}^{(s+1)} \leftarrow \mathbf{u}^k + \Delta t \sum_{n=0}^s \beta_{s+1,n} \dot{\mathbf{u}}^{(n)}$ 
12: end for
13:  $\mathbf{r}^{k+1} \leftarrow \mathbf{r}^{(S)}$ 
14:  $\mathbf{u}^{k+1} \leftarrow \mathbf{u}^{(S)}$ 

```

Here, $\delta \mathbf{u}$ denotes the shifting velocity associated with particle shifting, which is set to $\mathbf{0}$ when no particle shifting is applied. It should be noted that the Eulerian framework can also be obtained by setting $\delta \mathbf{u} = -\mathbf{u}$, ensuring that $\dot{\mathbf{r}} = \mathbf{0}$.

The new form of the pressure Poisson equation, Eq. (75), solved in line 7, is central to the proposed time-marching schemes for attaining high-order accuracy. Replacing it with other forms can significantly degrade the accuracy of the incompressibility condition, thereby deteriorating the convergence order of both velocity and pressure. Another advantage of the proposed schemes is their generality for arbitrary particle movement. Different frameworks, such as Eulerian, Lagrangian, and quasi-Lagrangian, can be accommodated solely through the definition of $\delta \mathbf{u}$. Moreover, the additional step for particle shifting, required in the conventional time-marching schemes, is eliminated. This is advantageous in terms of computational efficiency. Furthermore, compared to the split-type scheme described in Section 3.4, numerical stability is improved by avoiding the explicit convection term in the provisional velocity. The convective effects are instead captured through $-\nabla \mathbf{u}^{(s)} : \nabla \mathbf{u}^{(s)} + (\dot{\mathbf{r}}^{(s)} - \mathbf{u}^{(s)}) \cdot \nabla \nabla \cdot \mathbf{u}^{(s)}$ in line 7 and $(\dot{\mathbf{r}}^{(s)} - \mathbf{u}^{(s)}) \cdot \nabla \mathbf{u}^{(s)}$ in line 8. A comparative study of numerical stability is presented in Section 5.5.

5. Numerical results

This section presents numerical simulations of incompressible flows to investigate the validity of the proposed time-marching schemes (RK1–RK4). These schemes are expected to achieve convergence orders ranging from first order to fourth order for both velocity and pressure. In the Lagrangian case (i.e., without particle shifting), particle position is also expected to exhibit the same order. To determine the temporal accuracy, errors arising from other factors must be sufficiently suppressed. For this purpose, a

fourth-order spatial discretization scheme is employed as needed. In addition, the accuracy of the boundary conditions is critical, particularly that of the wall boundary condition, which may limit overall accuracy [28]. To focus on temporal accuracy, test cases without wall boundaries are considered.

To determine the characteristics of conventional schemes and compare them with those of the proposed schemes, the following four time-marching schemes are also evaluated:

- MPS-type scheme: see Section 3.2,
- ISPH-type scheme: see Section 3.3,
- Simplified ISPH-type scheme: identical to ISPH-type scheme, but without provisional particle movement [10],
- Split-type scheme: see Section 3.4.

In accordance with their original frameworks, the MPS-type, ISPH-type, and Simplified ISPH-type schemes are applied exclusively to Lagrangian cases. In contrast, the Split-type and proposed schemes are tested for Lagrangian, Eulerian, and quasi-Lagrangian cases.

5.1. Rigid body motion of a droplet

The first test case examines the rigid body motion of a circular droplet, which is a fundamental test for Lagrangian incompressible free-surface flow solvers. A droplet of radius R , initially centered at the origin, is assigned a velocity distribution corresponding to rigid body motion, defined by a uniform translational velocity U and an angular velocity ω as follows:

$$\mathbf{u}(\mathbf{x}, t = 0) = \begin{bmatrix} U - \omega y \\ \omega x \end{bmatrix}. \quad (79)$$

Droplet motion in the absence of gravity is simulated with the following parameters: $\rho = 1 \text{ kg/m}^3$, $\nu = 0.01 \text{ m}^2/\text{s}$, $R = 1 \text{ m}$, $U = 1 \text{ m/s}$, $\omega = 1 \text{ rad/s}$, and without particle shifting. Under these conditions, the droplet undergoes flow instability due to centrifugal forces. The objective of this test case is to evaluate how accurately numerical methods preserve the initial rigid body motion. The simulation time is restricted to $t \leq 5 \text{ s}$, which is well before the onset of instability.

The analytical solutions for the Lagrangian particle movement, flow velocity, and pressure are given by

$$\mathbf{r}(t) = \begin{bmatrix} \cos(\omega t) & -\sin(\omega t) \\ \sin(\omega t) & \cos(\omega t) \end{bmatrix} \mathbf{r}(0) + \begin{bmatrix} U t \\ 0 \end{bmatrix}, \quad \mathbf{u}(\mathbf{x}, t) = \begin{bmatrix} U - \omega y \\ \omega(x - U t) \end{bmatrix}, \quad P(\mathbf{x}, t) = \frac{\rho \omega^2}{2} ((x - U t)^2 + y^2 - R^2). \quad (80)$$

Velocity and pressure are linear and quadratic functions of position $\mathbf{x}$. Therefore, the second-order LSMPS scheme is employed because it can exactly reproduce these functions without introducing spatial truncation errors. As a result, numerical errors arising solely from

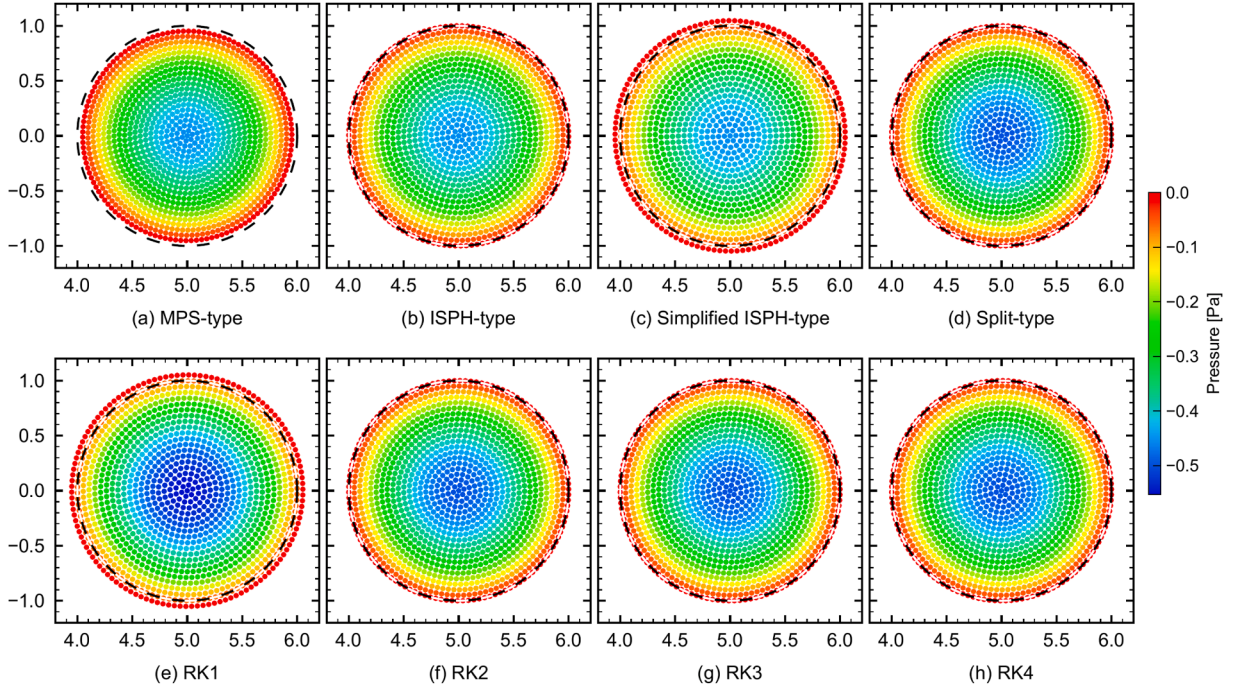


Fig. 1. Snapshots of simulation results at $t = 5 \text{ s}$, computed with $\Delta x = R/20$ (1309 particles) and $\Delta t = 0.025 \text{ s}$, for the rigid body motion of a droplet. Dashed lines indicate analytical free-surface positions.

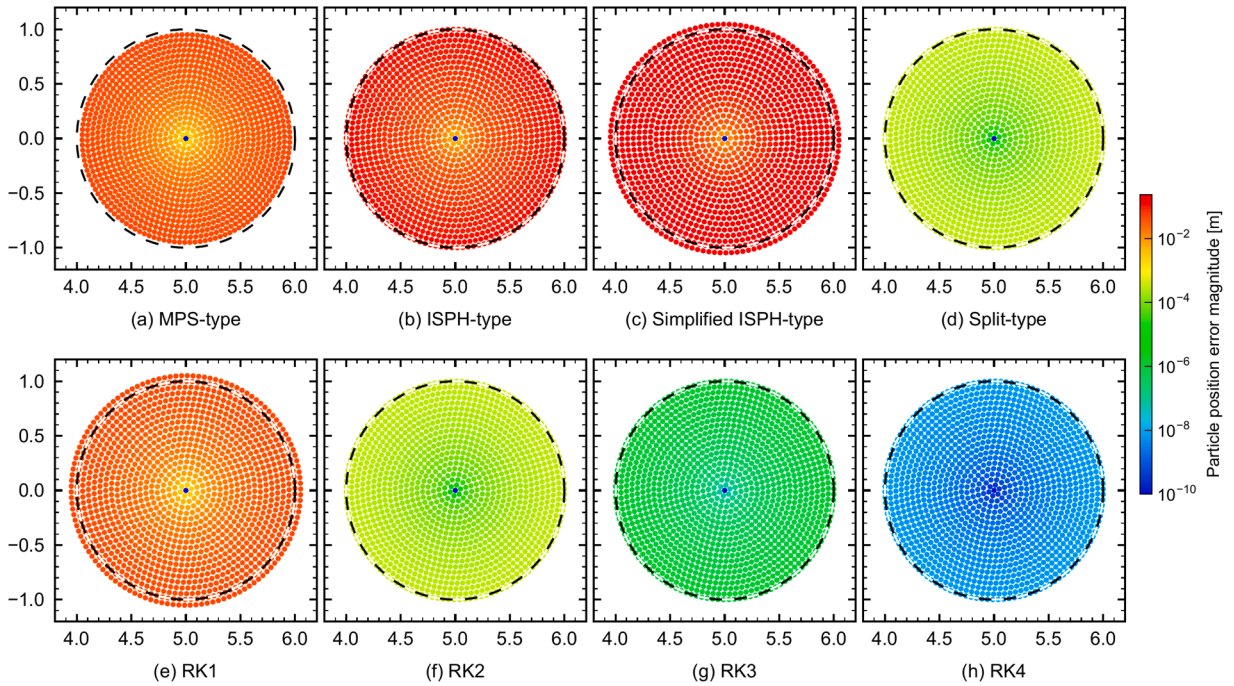


Fig. 2. Distributions of particle position error magnitude at $t = 5$ s for the rigid body motion of a droplet.

the time-marching scheme can be observed. Because the accuracy is independent of the spatial resolution, the characteristic particle spacing is fixed at $\Delta x = R/20$ (1309 particles) throughout this test case, while the time step interval Δt is varied to evaluate temporal convergence.

In Fig. 1, snapshots of the simulation results at $t = 5$ s, computed with $\Delta t = 0.025$ s, are displayed for each time-marching scheme. All results yield similar pressure distributions; however, notable differences can be observed in droplet radius, primarily due to variations in the computation of particle positions. The MPS-type scheme results in shrinking of the droplet as a result of first-order backward time integration, whereas the RK1 scheme results in droplet expansion due to first-order forward time integration. The other schemes effectively preserve the initial radius, with the exception of the Simplified ISPH-type scheme. Although its formulation is second-order due to the use of the midpoint velocity, the omission of particle movement reduces its overall accuracy compared to that of the ISPH-type scheme.

Fig. 2 presents the distributions of the particle position error magnitude. Notably, the ISPH-type scheme exhibits a large error magnitude despite maintaining an accurate droplet radius. This error is due to the reduced rotational speed: at $t = 5$ s, the angular velocity uniformly decreases by approximately 5 % compared to the initial value, as illustrated in Fig. 3. The same behavior is observed for the Simplified ISPH-type scheme, which exhibits an angular velocity decrease of approximately 9 %. These results are markedly worse than those of all other schemes, whose errors are below 10^{-12} %.

Fig. 4 presents the temporal convergence of the error norms for instantaneous physical quantities at $t = 5$ s, computed for various time step intervals Δt . The error norms for particle position, flow velocity, and pressure are defined as the ℓ^2 -norms of the nodal errors, normalized by the ℓ^2 -norms of the corresponding analytical solutions. The velocity divergence is computed from the resulting instantaneous velocity field. Almost all schemes exhibit at least the expected convergence orders, with two exceptions—the ISPH-type and Simplified ISPH-type schemes—which exhibit only first-order convergence for particle position due to the inaccurate velocity field. All other schemes yield highly accurate velocity and velocity divergence, with only round-off errors. For particle position, the MPS-type and RK1 schemes exhibit first-order convergence, while the Split-type and RK2 schemes exhibit second-order convergence. Lastly, the RK3 and RK4 schemes exhibit third- and fourth-order convergence, respectively. These results are consistent with theoretical predictions, as the power series expansion of the particle movement in Eq. (80) yields the following complete polynomials:

$$x(t + \Delta t) = x(t) + (U - \omega y)\Delta t - \frac{1}{2}\omega^2 x \Delta t^2 + \frac{1}{6}\omega^3 y \Delta t^3 + \frac{1}{24}\omega^4 x \Delta t^4 - \frac{1}{120}\omega^5 y \Delta t^5 - \frac{1}{720}\omega^6 x \Delta t^6 + \dots, \quad (81)$$

$$y(t + \Delta t) = y(t) + \omega x \Delta t - \frac{1}{2}\omega^2 y \Delta t^2 - \frac{1}{6}\omega^3 x \Delta t^3 + \frac{1}{24}\omega^4 y \Delta t^4 + \frac{1}{120}\omega^5 x \Delta t^5 - \frac{1}{720}\omega^6 y \Delta t^6 + \dots. \quad (82)$$

The errors in the pressure field are due to errors in the angular velocity and droplet radius. According to Eqs. (81) and (82), the local truncation error of the squared droplet radius R^2 can be demonstrated to be a polynomial of even degree. Therefore, the observed odd-order convergence of the pressure error norm is also consistent with theoretical predictions.

The results of this test case indicate that the proposed time-marching schemes achieve high-order temporal accuracy, at least for particle position. The ISPH-type and Simplified ISPH-type schemes demonstrate a critical limitation in the conservation of rotational

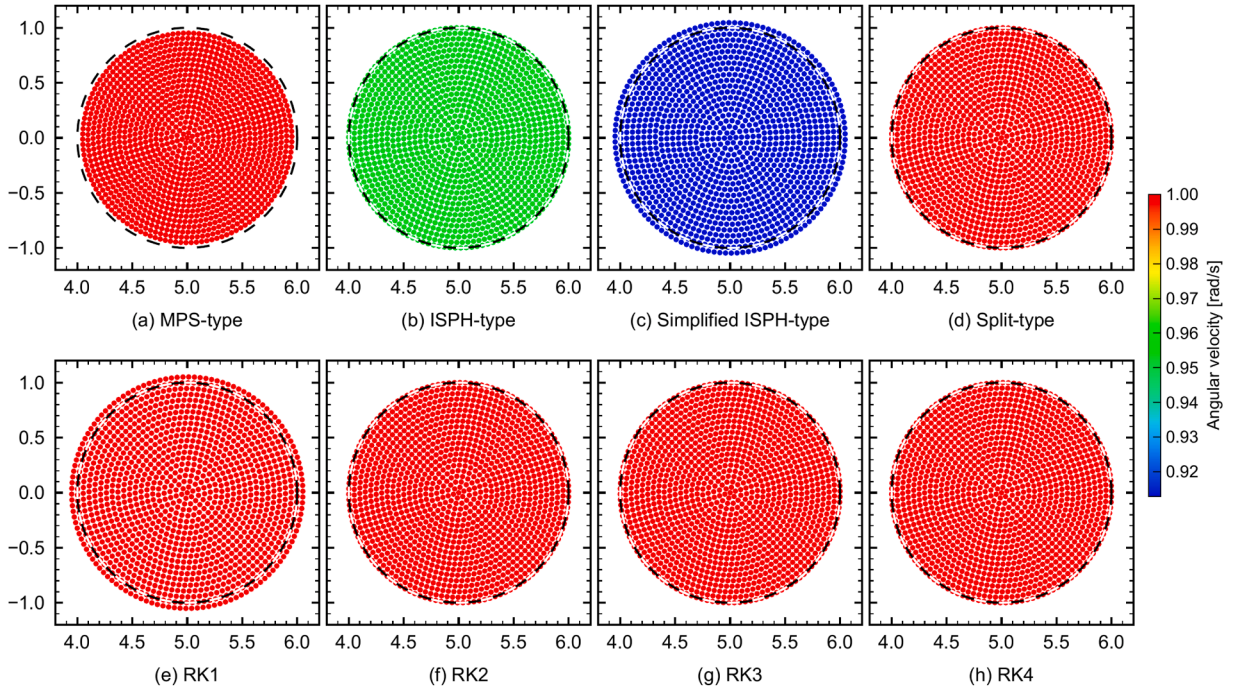


Fig. 3. Distributions of angular velocity at $t = 5$ s for the rigid body motion of a droplet. The ground truth is uniformly 1 rad/s.

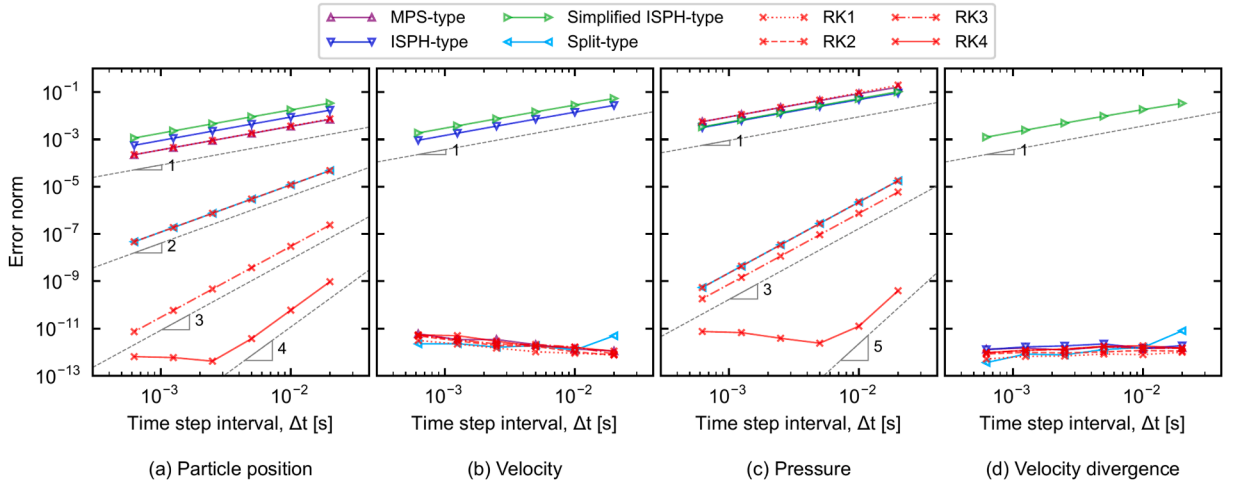


Fig. 4. Temporal convergence of the instantaneous error norms at $t = 5$ s for the rigid body motion of a droplet. Simulations are performed with a constant particle spacing $\Delta x = R/20$ and varying time step interval Δt .

motion. Furthermore, the Simplified ISPH-type scheme exhibits particularly large errors, demonstrating the importance of provisional particle movement in the ISPH-type scheme. It should also be noted that the Split-type scheme results in second-order convergence; however, this behavior is dependent on the specific problem.

5.2. Oscillating droplet under a central force field

The second test case examines an oscillating droplet under a central force field, which is a widely used benchmark for evaluating numerical stability and conservation properties [7,16,29–31]. Under a central force field $\mathbf{g} = -B^2\mathbf{x}$, an inviscid droplet with an initial radius R_0 undergoes periodic motion described by

$$\mathbf{u}(\mathbf{x}, t) = \begin{bmatrix} A(t)x \\ -A(t)y \end{bmatrix}, \quad (83)$$

where B and $A_0 := A(t=0)$ are input parameters that define the oscillation period and the maximum stretching of the droplet. The time evolution of $A(t)$ can be accurately determined by numerical time integration of the following equations:

$$\frac{d}{dt}A = \frac{(A^2 + B^2)(R_y^2 - R_x^2)}{R_x^2 + R_y^2}, \quad \frac{d}{dt}R_x = AR_x, \quad \frac{d}{dt}R_y = -AR_y, \quad (84)$$

with initial conditions $A(0) = A_0$, $R_x(0) = R_0$, and $R_y(0) = R_0$. Here, R_x and R_y denote the semi-axis lengths of the droplet in the x and y directions, respectively. The analytical solutions for the Lagrangian particle movement and pressure field are given by

$$\mathbf{r}(t) = \begin{bmatrix} R_x & 0 \\ 0 & R_y \end{bmatrix} \mathbf{r}(0), \quad P(\mathbf{x}, t) = \frac{\rho(A^2 + B^2)R_x^2 R_y^2}{R_x^2 + R_y^2} \left(1 - \frac{x^2}{R_x^2} - \frac{y^2}{R_y^2} \right). \quad (85)$$

Due to conservation of volume, $R_x R_y = R_0^2$ holds. Long-term simulations are performed with the following parameters: $\rho = 1 \text{ kg/m}^3$, $B = 3 \text{ s}^{-1}$, $A_0 = 1 \text{ s}^{-1}$, and $R_0 = 1 \text{ m}$. With these parameters, the oscillation period and the maximum and minimum stretching factors are $T \approx 1.50052 \text{ s}$, $R_{\max}/R_0 \approx 1.26310$, and $R_{\min}/R_0 \approx 0.791700$, respectively. As in the previous test case, the second-order LSMPS scheme and a constant particle spacing of $\Delta x = R_0/20$ are employed. Simulations are performed without particle shifting, and the time step interval Δt is varied to assess temporal convergence.

Fig. 5 presents snapshots of the simulation results at $t = 1.75T$, computed with $\Delta t = T/100$, for each time-marching scheme. At this time point, the droplet has its maximum elongation in the y direction. The RK1 and Split-type schemes overestimate droplet deformation, whereas the MPS-type, ISPH-type, and Simplified ISPH-type schemes underestimate it. In contrast, the RK2, RK3, and RK4 schemes exhibit good agreement with the semi-analytical solution obtained by Eqs. (84) and (85).

Fig. 6 illustrates the temporal convergence of the instantaneous error norms at $t = 0.9T$. The definitions of the error norms are the same as those in the previous test case. Notably, the Simplified ISPH-type scheme exhibits high accuracy for particle position and velocity compared to the other conventional schemes; however, its considerably large velocity divergence error indicates its fundamental limitation as an incompressible flow solver. The convergence behavior of the other schemes is in perfect agreement with the expected results. For particle position, velocity, and pressure, the conventional MPS-type, ISPH-type, and Split-type schemes exhibit first-order convergence. The proposed RK1–RK4 schemes exhibit first- to fourth-order convergence, respectively. In addition,

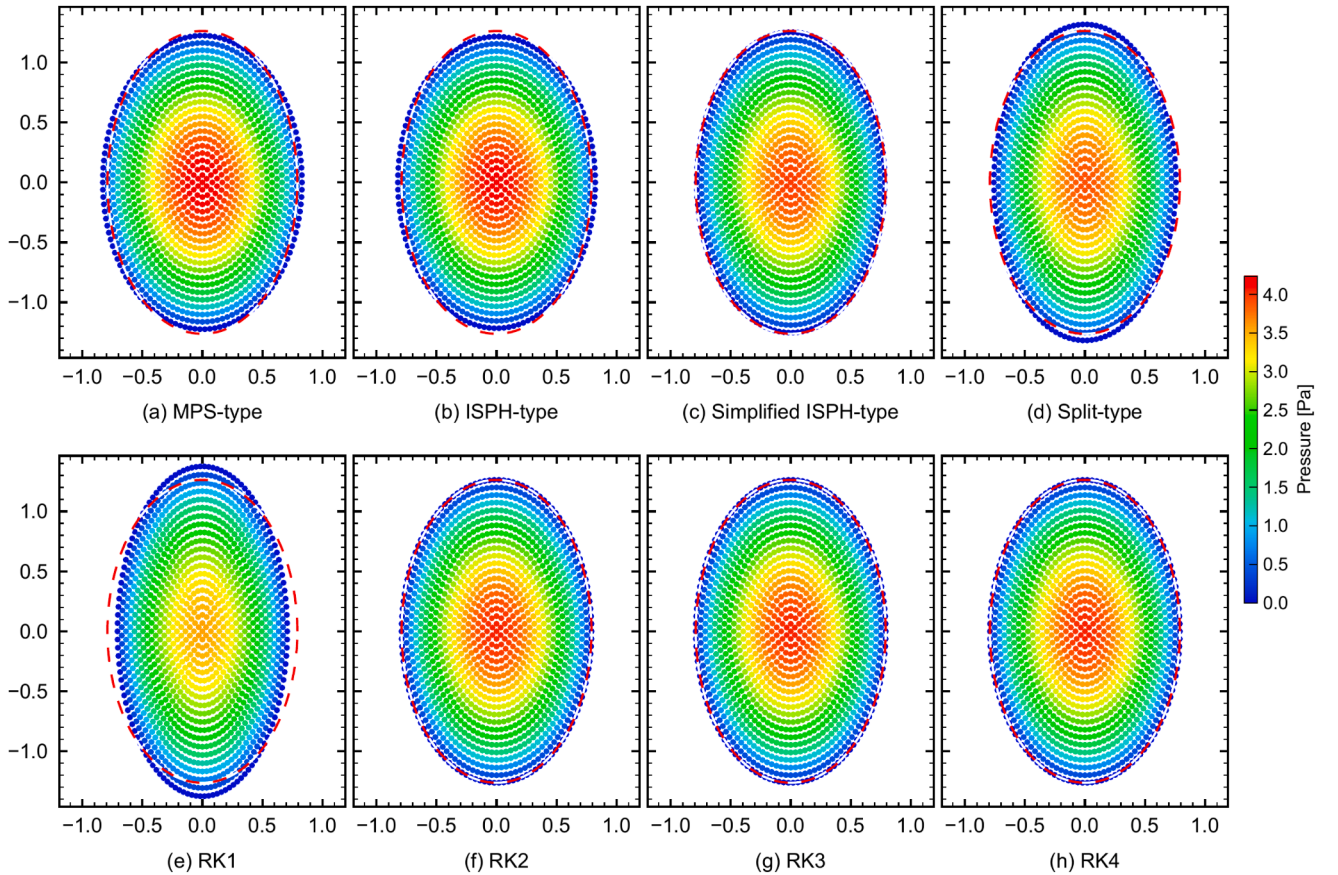


Fig. 5. Snapshots of simulation results at $t = 1.75T$, computed with $\Delta x = R_0/20$ (1309 particles) and $\Delta t = T/100$, for an oscillating droplet. Dashed lines indicate analytical free-surface positions.

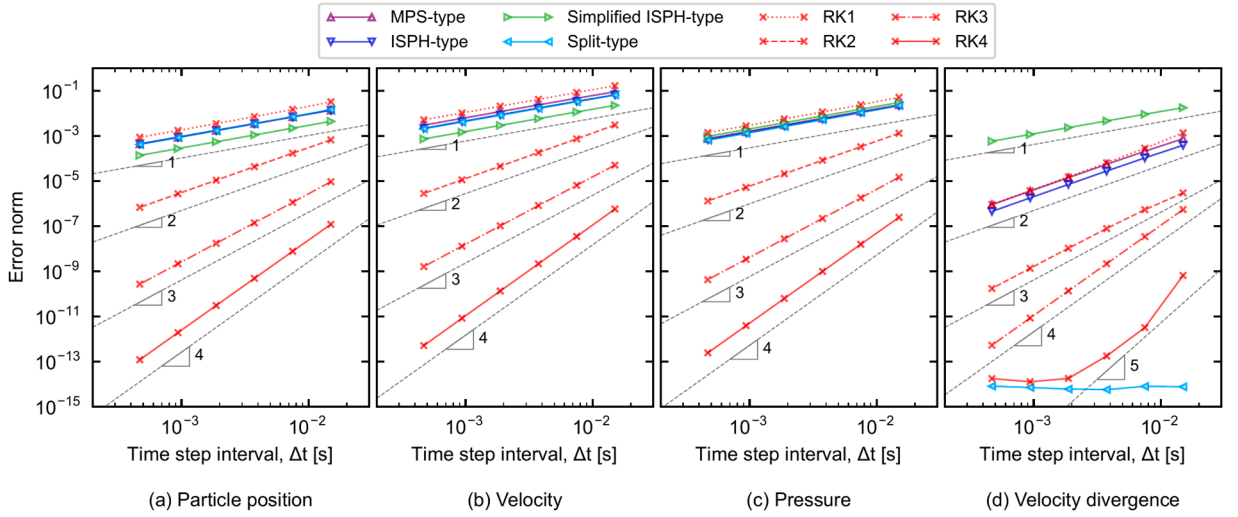


Fig. 6. Temporal convergence of instantaneous error norms at $t = 0.9T$ for an oscillating droplet. Simulations are performed with a constant particle spacing $\Delta x = R/20$ and varying time step interval Δt .

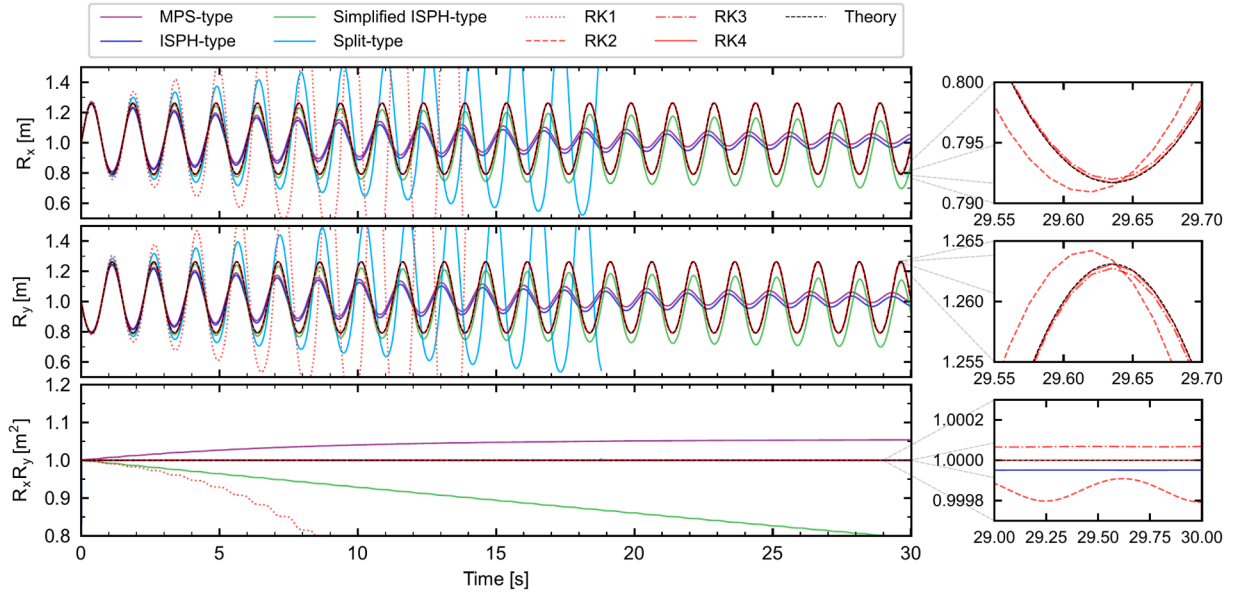


Fig. 7. Time histories of the semi-axis lengths (R_x, R_y) of the droplet and their product ($R_x R_y$) in the time range $0 \leq t \leq 20T$ for an oscillating droplet. Simulations are performed with $(\Delta x, \Delta t) = (R_0/20, T/100)$, where $R_0 = 1$ m and $T \approx 1.50052$ s.

the convergence rate for velocity divergence is one order higher than for that for the other quantities, which is consistent with the theoretical predictions.

Fig. 7 presents the time histories of the semi-axis lengths of the droplet and their product for $0 \leq t \leq 20T$, with simulations performed using $\Delta t = T/100$. The RK1 and Split-type schemes diverge in the middle of the simulation due to the increasing amplitude of the oscillation. In contrast, the MPS-type and ISPH-type schemes produce stable solutions, although the oscillations are numerically damped with time. The RK2, RK3, and RK4 schemes demonstrate excellent overall agreement with the semi-analytical solutions and preserve the product $R_x R_y$, indicating accurate volume conservation.

Fig. 8 plots the time histories of the kinetic, potential, and total energies, defined as follows:

$$E_k(t) = \frac{1}{2N} \sum_{i=1}^N \|u_i(t)\|^2, \quad E_p(t) = \frac{B^2}{2N} \sum_{i=1}^N \|r_i(t)\|^2, \quad E_{\text{total}}(t) = E_k(t) + E_p(t), \quad (86)$$

where N denotes the number of particles. The divergent behavior of the RK1 and Split-type schemes and the numerical damping of the MPS-type and ISPH-type schemes are observed from the kinetic energy. Although the Simplified ISPH-type scheme produces

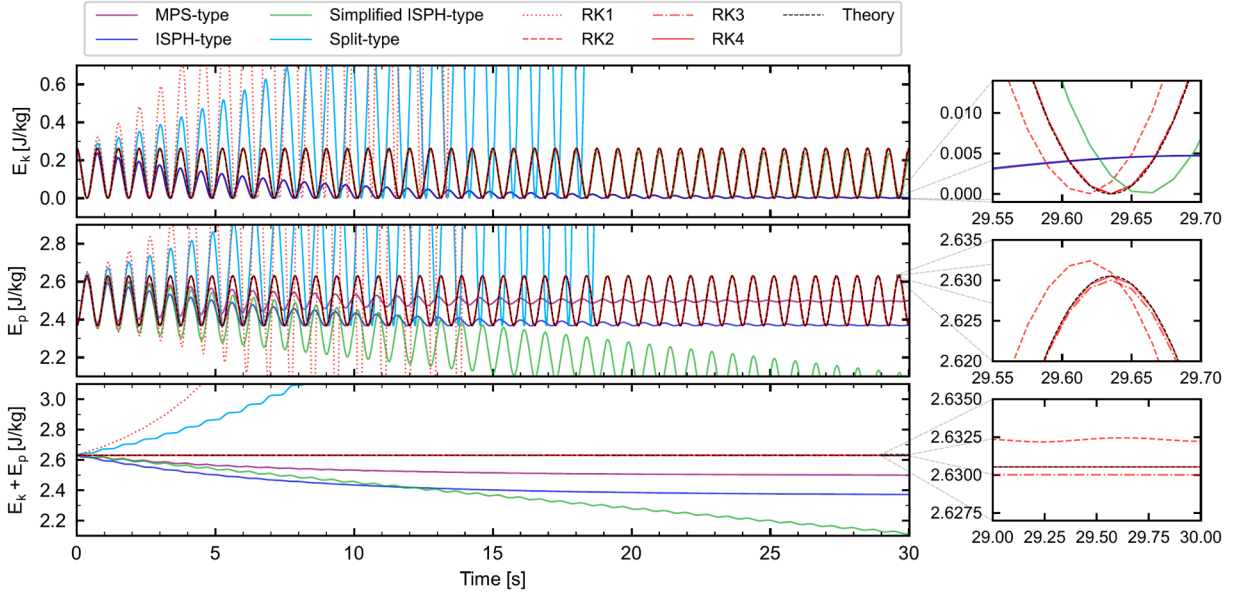


Fig. 8. Time histories of kinetic (E_k), potential (E_p), and total ($E_k + E_p$) energies in the time range $0 \leq t \leq 20T$ for an oscillating droplet. The simulation conditions are the same as those in Fig. 7.

accurate results for kinetic energy, the potential energy decreases considerably due to droplet volume reduction. In contrast, the RK2, RK3, and RK4 schemes are in good agreement with the theory for both kinetic and potential energies, resulting in highly accurate conservation of total energy.

The above results demonstrate that the proposed high-order schemes (RK2, RK3, and RK4) successfully achieve high-order temporal accuracy. Notably, even the second-order RK2 scheme substantially improves accuracy compared to first-order schemes. Furthermore, this increased accuracy contributes to improved conservation of volume and energy.

5.3. Taylor–Green vortex

The Taylor–Green vortex problem is simulated to evaluate the accuracy of the proposed time-marching schemes for incompressible flow with complex velocity fields. The computational domain is defined as $-L \leq x \leq L$, $-L \leq y \leq L$, with periodic boundary conditions. The initial velocity distribution and the analytical solutions for the velocity and pressure fields are given by

$$\mathbf{u}(\mathbf{x}, t) = U \exp\left(-\frac{2\pi^2 \nu t}{L^2}\right) \begin{bmatrix} \sin(\pi x/L) \cos(\pi y/L) \\ -\cos(\pi x/L) \sin(\pi y/L) \end{bmatrix}, \quad (87)$$

$$P(\mathbf{x}, t) = \frac{\rho U^2}{4} \exp\left(-\frac{4\pi^2 \nu t}{L^2}\right) (\cos(2\pi x/L) + \cos(2\pi y/L)). \quad (88)$$

The simulation parameters are set as $L = 1$ m, $U = 1$ m/s, $\rho = 1$ kg/m³, and $\nu = 10^{-3}$ m²/s, yielding a Reynolds number of $Re = LU/\nu = 10^3$. Due to the flow structure, the Lagrangian particle movement generates highly distorted particle distributions, leading to numerical instabilities. Therefore, only the Split-type scheme and the proposed time-marching schemes are tested under Eulerian and quasi-Lagrangian conditions, and their performances are compared. To minimize spatial discretization errors, the fourth-order LSMPs scheme is employed.

Fig. 9 presents the distributions of the velocity error magnitude at $t = 0.5$ s for each time-marching scheme in both Eulerian and quasi-Lagrangian frameworks. The simulations are performed with $\Delta x = L/64$ (128×128 particles) and $\Delta t = 0.5/160$ s. In the Eulerian framework, the Split-type and RK1 schemes exhibit similar error magnitudes overall, whereas the RK2, RK3, and RK4 schemes exhibit noticeably smaller errors. In the quasi-Lagrangian framework, all schemes exhibit larger errors than in the Eulerian case due to errors associated with particle movement. However, the increase in error magnitude is not uniform across the schemes. The Split-type scheme effectively suppresses the error increase, demonstrating the effectiveness of the operator splitting approach. In contrast, the RK1 scheme exhibits error magnitudes several orders higher in the quasi-Lagrangian framework than in the Eulerian framework. Notably, as the scheme order increases, the error magnitude is considerably reduced, with the RK3 and RK4 schemes exhibiting error magnitudes similar to those in the Eulerian case.

Fig. 10 presents a comparison of the instantaneous error norms at $t = 0.5$ s, where the particle spacing Δx and time step interval Δt are varied as the Courant number is fixed at $U\Delta t/\Delta x = 0.2$. The results indicate that the magnitude of the error norm is strongly affected by the choice of time-marching scheme and framework, particularly at higher resolutions, indicating characteristic differences in temporal accuracy. Overall, the Eulerian framework exhibits higher accuracy and better convergence than the quasi-Lagrangian

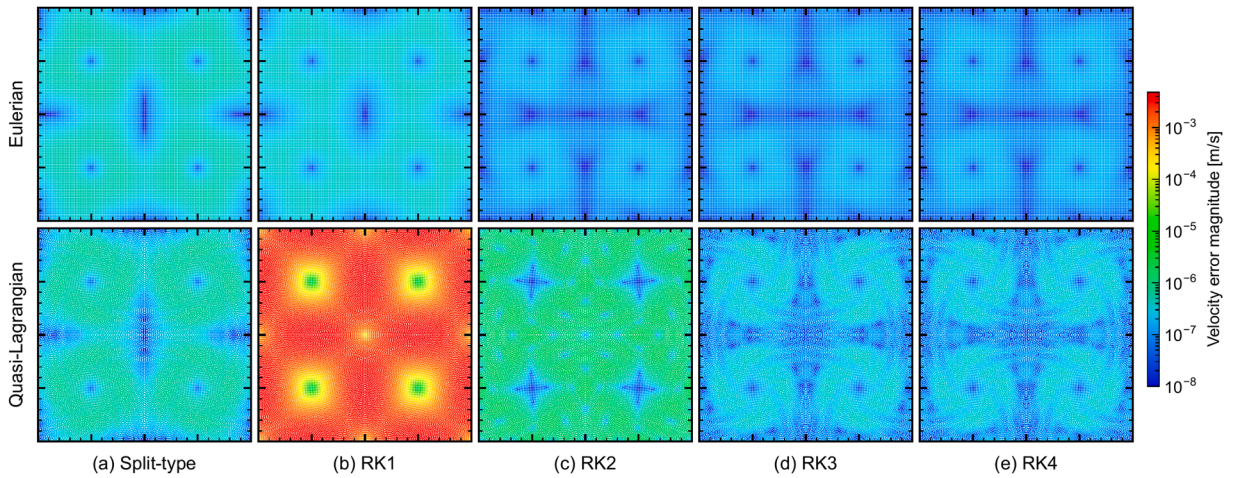


Fig. 9. Distributions of velocity error magnitude in the domain $-L \leq x \leq L$ and $-L \leq y \leq L$ at $t = 0.5$ s, computed with $\Delta x = L/64$ (128^2 particles) and $\Delta t = 0.5/160$ s, for the Taylor–Green vortex.

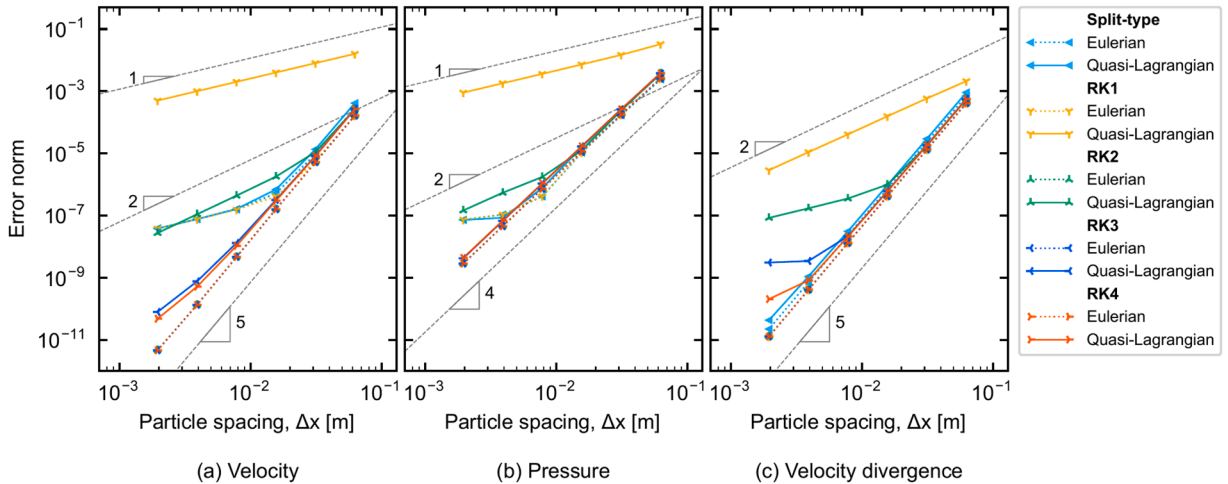


Fig. 10. Convergence of instantaneous error norms at $t = 0.5$ s for the Taylor–Green vortex. Simulations are performed with varying particle spacing Δx and time step interval $\Delta t = 0.2\Delta x/U$.

framework for all schemes. Among them, the Split-type scheme exhibits a relatively small decrease in accuracy, despite its temporal accuracy being only first order. The poorest result is observed for the RK1 scheme in the quasi-Lagrangian framework, which exhibits a substantial increase in error norms for all quantities. The RK2 scheme demonstrates a similar tendency, but its second-order convergence results in a smaller velocity error norm than that of the Split-type scheme at the highest resolution, demonstrating the advantage of higher-order schemes. The RK3 and RK4 schemes achieve the highest accuracy for velocity and pressure.

In this test case, the performance of the Split-type scheme is particularly notable. This result is mainly attributed to the fact that the vortices evolve primarily through viscous diffusion, while their centers remain at the initial locations. Under such conditions, the magnitude of the acceleration in the Eulerian description, $\|\partial \mathbf{u} / \partial t\|$, is generally much smaller than that in the Lagrangian description, $\|D\mathbf{u} / Dt\|$. The Split-type scheme is designed to leverage this property: the time increment is computed using the local acceleration, and the particle movement is determined using a frozen velocity field. However, this scheme also has critical limitations, including stability problems and a lack of Galilean invariance, as discussed in an earlier section.

5.4. Merging of two co-rotating vortices

The merging of two co-rotating Lamb–Oseen vortices is simulated to evaluate the performance of the proposed time-marching schemes for more practical problems. The velocity and vorticity distributions of a single two-dimensional Lamb–Oseen vortex are

defined as

$$\mathbf{u}_{\text{LO}}(\mathbf{x}) = \frac{\Gamma}{2\pi\|\mathbf{x}\|^2} \left(1 - \exp\left(-\frac{\|\mathbf{x}\|^2}{a_0^2}\right) \right) \begin{bmatrix} -y \\ x \end{bmatrix}, \quad \omega_{\text{LO}}(\mathbf{x}) = \frac{\Gamma}{\pi a_0^2} \exp\left(-\frac{\|\mathbf{x}\|^2}{a_0^2}\right), \quad (89)$$

where Γ and a_0 denote the circulation and initial core radius, respectively. The initial positions of the vortices are set as $\mathbf{x}_1 = (-b_0/2, 0)$ and $\mathbf{x}_2 = (b_0/2, 0)$, where b_0 denotes the initial separation distance. The initial velocity field is set as

$$\mathbf{u}(\mathbf{x}, t=0) = \alpha(\|\mathbf{x}\|, r_1, r_2) (\mathbf{u}_{\text{LO}}(\mathbf{x} - \mathbf{x}_1) + \mathbf{u}_{\text{LO}}(\mathbf{x} - \mathbf{x}_2)), \quad (90)$$

where $\alpha(r, r_1, r_2)$ is a smooth function defined by

$$\alpha(r, r_1, r_2) = \begin{cases} 1 & (r < r_1) \\ 1 - 3\left(\frac{r-r_1}{r_2-r_1}\right)^2 + 2\left(\frac{r-r_1}{r_2-r_1}\right)^3 & (r_1 \leq r \leq r_2) \\ 0 & (r_2 < r) \end{cases} \quad (91)$$

The computational domain is defined as $-10a_0 \leq x \leq 10a_0$ and $-10a_0 \leq y \leq 10a_0$ with periodic boundaries and $(r_1, r_2) = (8a_0, 10a_0)$. The simulation parameters are set as $\Gamma = 1 \text{ m}^2/\text{s}$, $a_0 = 1 \text{ m}$, $b_0 = 5a_0$, $\rho = 1 \text{ kg/m}^3$, and $\nu = 10^{-3} \text{ m}^2/\text{s}$, resulting in $Re = \Gamma/\nu = 10^3$. The fourth-order LSMPs scheme is employed. Simulations are performed in the quasi-Lagrangian framework with different Δx and Δt to determine the resolution dependence, maintaining a fixed ratio of $\Delta t/\Delta x = 2.5a_0/\Gamma$. For comparison, reference solutions are obtained using the open-source FVM software OpenFOAM v2412. FVM simulations are performed using the second-order upwind divergence (convection) scheme and the backward Euler time integration scheme at high spatial and temporal resolutions, with the ratio $\Delta t/\Delta x$ held constant in the same manner.

Before presenting the simulation results, the vortex merging process is briefly described. The dynamics of co-rotating vortex pairs have been extensively studied [32–35]. When the separation distance b is sufficiently large compared to the core radius a , the vortices rotate around each other at a constant angular speed, $\Omega = \Gamma/(\pi b_0^2)$, while the core radius increases due to viscous diffusion according to $a^2 = a_0^2 + 4\nu t$. The vortices start to merge when the core radius a exceeds a critical fraction of the separation distance b . This critical ratio, $(a/b)_c$, has been investigated in numerous studies. For example, an experimental study [32] reported $(a/b)_c \approx 0.24$. Using this value, the critical time t_c for the present simulation can be estimated as $t_c = ((a/b)_c^2 - (a/b_0)^2) b_0^2/(4\nu) \approx 110 \text{ s}$. After the onset of merging ($t \gtrsim t_c$), the vortices begin to deform considerably, and the separation distance decreases until b/b_0 reaches approximately 0.25. This convective merging phase is central to the overall vortex merging process. It is followed by a diffusive merging phase, in which the separation distance continues to decrease, accompanied by some perturbations, and the vortices eventually merge into a single vortex.

Fig. 11 presents snapshots of the simulation results at several time points from $t = 50 \text{ s}$ (prior to the onset of merging) to $t = 500 \text{ s}$ (after the completion of merging). The vorticity fields and trajectories of the vortex centers, defined by the locations of vorticity maxima, are depicted for the FVM, Split-type, and RK2 schemes. The spatial and temporal resolutions for the FVM simulation are $\Delta x = a_0/80$ (1600×1600 cells) and $\Delta t = 1/32 \text{ s}$, respectively, while those for the particle method with the Split-type and RK2 schemes are $\Delta x = a_0/10$ (200×200 particles) and $\Delta t = 1/4 \text{ s}$, respectively. Overall, the results are similar, but notable differences are observed for the Split-type scheme, particularly at $t = 350 \text{ s}$. The Split-type scheme exhibits a slightly larger vorticity maximum and a delayed rotation of the vortex cores compared to the FVM results. In contrast, the RK2 scheme exhibits good agreement with the reference FVM solution throughout the simulation period. The discrepancy between the Split-type and RK2 schemes arises purely from temporal errors since both schemes use the same spatial discretization. This result indicates the limited accuracy of the Split-type scheme in simulating the convective merging phase.

Fig. 12 displays time histories of the maximum vorticity and the separation distance between the vortex centers at three different resolutions. Overall, the RK1 and Split-type schemes exhibit the worst performance, producing considerably larger maximum vorticity and separation distances than the other schemes. In contrast, the RK2, RK3, and RK4 schemes yield values much closer to those obtained by FVM. Moreover, although the resolution dependence of FVM is fairly small, that of the high-order proposed schemes is even smaller, with almost identical results at the two highest resolutions. This suggests that the proposed schemes yield more accurate solutions than FVM in the present case. Furthermore, the results indicate that temporal accuracy is critical for simulating the vortex merging process.

Lastly, Table 2 summarizes the computation times² required to simulate the merging of two co-rotating vortices for $0 \leq t \leq 500 \text{ s}$. Wall-clock times were measured on a workstation equipped with two Intel Xeon Gold 6136 CPUs. At the same resolution, the particle method requires approximately five times more computation time than FVM. For the different time-marching schemes used in the particle method, computation time is found to scale approximately linearly with the number of stages. This result is expected because the dominant computational cost is associated with solving the pressure Poisson equation. Because the RK2, RK3, and RK4 schemes achieve considerably higher temporal accuracy even at lower resolutions, they outperform the Split-type scheme and the RK1 scheme in terms of computational efficiency. Furthermore, these high-order schemes demonstrate superior performance compared to FVM under the current conditions, although this advantage may diminish if a higher-order scheme is applied to FVM.

² The reported computation times are provided for reference only, as the current implementation is not fully optimized for performance. In addition, some irregularities are observed during MPI execution with the FVM software employed.

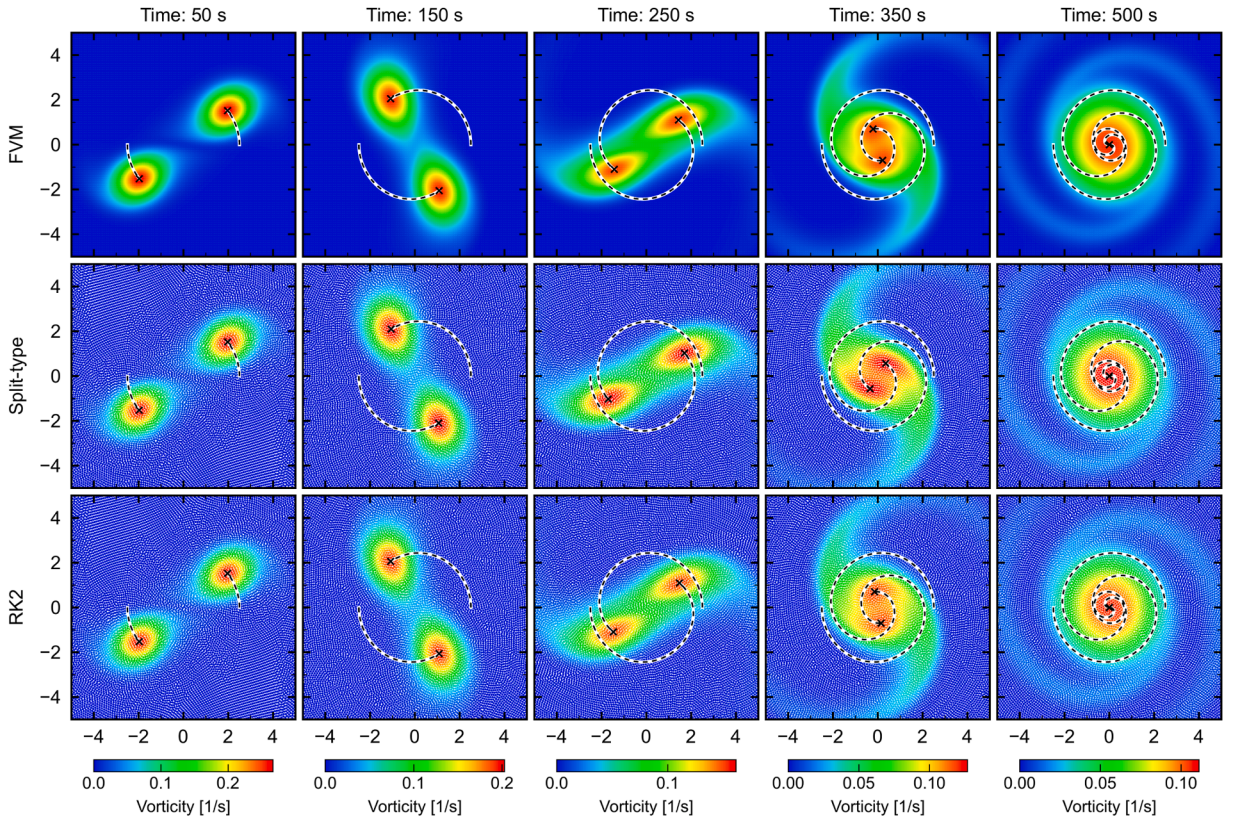


Fig. 11. Snapshots of simulation results for the merging of two co-rotating vortices. Simulations are performed with $(\Delta x, \Delta t) = (a_0/80, 1/32 \text{ s})$ for FVM and $(a_0/10, 1/4 \text{ s})$ for the Split-type and RK2 schemes, where $a_0 = 1 \text{ m}$. Dashed lines indicate the trajectories of the vortex centers, defined by the locations of vorticity maxima.

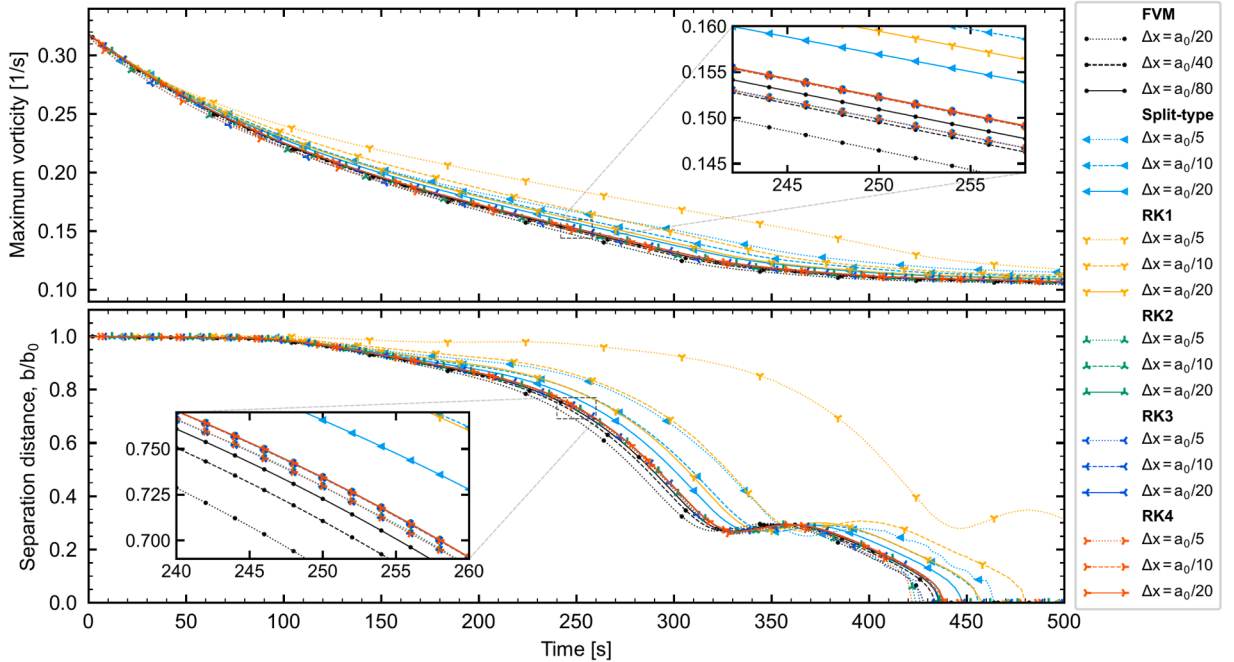


Fig. 12. Time histories of the maximum vorticity and the separation distance between two vortex centers during the merging of two co-rotating vortices.

Table 2

Computation times for simulating the merging of two co-rotating vortices. A workstation equipped with two Intel Xeon Gold 6136 CPUs was employed.

Δx	Δt	Number of cells/particles	Number of time steps	Wall-clock time (in seconds)					
				FVM*	Split-type	RK1	RK2	RK3	RK4
$a_0/5$	1/2 s	10 000	1 000	10	43	42	76	112	147
$a_0/10$	1/4 s	40 000	2 000	53	298	273	523	754	944
$a_0/20$	1/8 s	160 000	4 000	445	2 187	2 157	3 803	5 617	7 182
$\sim a_0/40$	1/16 s	640 000	8 000	7 109 [†]	—	—	—	—	—
$a_0/80$	1/32 s	2 560 000	16 000	284 794	—	—	—	—	—

* OpenFOAM v2412 was used. [†] Value possibly affected by a computational anomaly.

5.5. Collision of planar opposed jets

The collision of planar opposed jets is simulated to evaluate the applicability of the proposed time-marching schemes under large free-surface deformations. The liquid jets are assumed inviscid, and are vertically injected from inlets located at the upper and lower boundaries of the computational domain. The inlet conditions are set symmetrically with respect to the $y = 0$ plane. In this configuration, the jet behavior is essentially equivalent to that of an inviscid liquid jet impinged on a flat plate [4,14,36–39], and the same analytical solution applies.

The computational setup follows the previous study [4]. A uniform velocity U is imposed on the inlets at $y = \pm 3H$, where H denotes the inlet width. Outlet boundaries are placed at $x = \pm 20H$ to observe the stability in the downstream region. The simulation parameters are set as $H = 1$ m, $U = 1$ m/s, and $\rho = 1$ kg/m³. To prevent severe numerical instabilities and consequent simulation breakdown caused by the generation of splashes and the impact of colliding jets, the spatial discretization is performed using the second-order LSMPs scheme with stabilization techniques: regularized least squares and velocity-coupled pressure gradient calculation (see [4] for details). The Split-type scheme and the proposed RK2 scheme are compared under quasi-Lagrangian particle movement. Particle shifting is applied only to internal particles to capture free-surface deformations accurately. The spatial resolution and the time step interval are set to $\Delta x = H/20$ and $\Delta t = 0.0125$ s respectively, which gives a Courant number $U \Delta t / \Delta x = 0.25$.

Fig. 13 presents snapshots of the simulation results, illustrating the time evolution between $t = 5$ s and 50 s. While the overall jet behaviors are qualitatively similar for both schemes, notable differences appear in the downstream region. At first, to validate the computed pressure values and free-surface shapes, an enlarged view of the jet collision at $t = 50$ s is presented in Fig. 14. The pressure distribution around the stagnation point is nearly identical between the two schemes. The maximum pressure agrees well with the analytical value, $P_{\max} = \rho U^2 / 2 = 0.5$ Pa. The free-surface shapes also show excellent agreement with the analytical solution [37], confirming the capability of both schemes to capture large free-surface deformations.

In the downstream region, however, the Split-type scheme produces unphysical oscillations of the free surface, whereas the RK2 scheme yields a flat surface consistent with the analytical solution. Fig. 15 plots the convective terms: $-\mathbf{u} \cdot \nabla \mathbf{u}$ for the Split-type scheme and $(\mathbf{r} - \mathbf{u}) \cdot \nabla \mathbf{u}$ for the RK2 scheme. Since the velocity distribution becomes uniform in the far downstream, the convective term should decay. Instead, in the Split-type scheme, it grows stronger downstream, reaching values several orders of magnitude larger than those in the RK2 scheme. This clearly indicates numerical instabilities caused by the explicit convection term. In contrast, the proposed schemes evaluate the convective effects in a moving reference frame, which improves numerical stability.

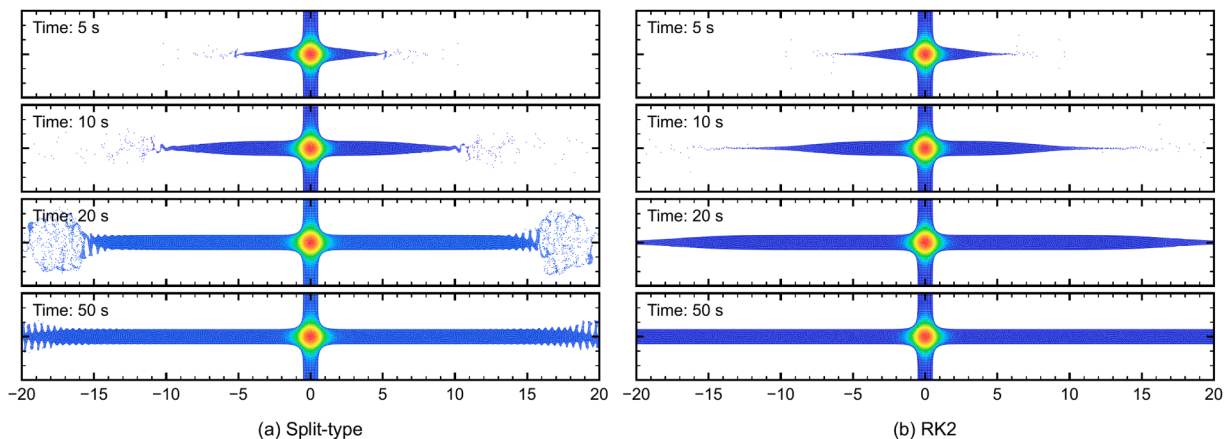


Fig. 13. Snapshots of simulation results for the collision of planar opposed jets. Color indicates pressure.

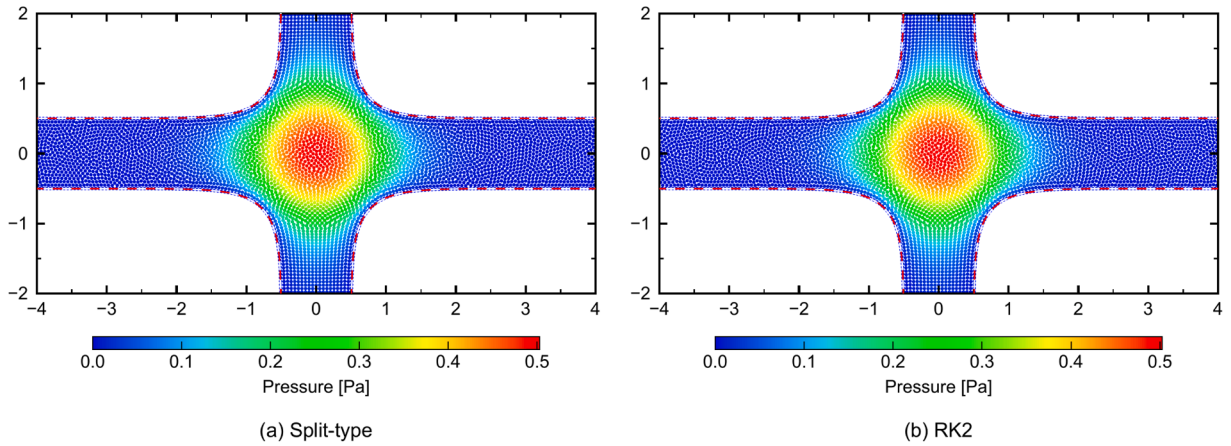


Fig. 14. Enlarged view of the pressure distribution around the stagnation point of planar opposed jets at $t = 50$ s. Dashed lines indicate analytical free-surface curves.

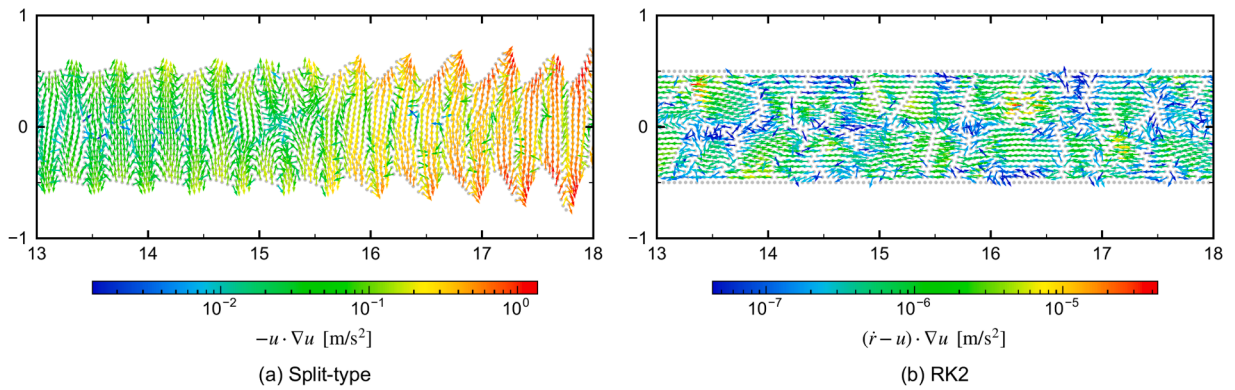


Fig. 15. Distributions of the convective terms in the downstream region of planar opposed jets at $t = 50$ s.

6. Conclusions

The primary contributions of this study are twofold: (i) the development of novel time-marching schemes that achieve high-order temporal accuracy for simulating incompressible flow in particle methods, and (ii) the clarification of the characteristics of both existing and proposed time-marching schemes through numerical investigations employing highly accurate spatial discretization.

In developing the novel time-marching schemes, formulations for the time evolution of computational variables along particle trajectories are introduced, resulting in the derivation of a new form of the pressure Poisson equation that explicitly accounts for particle movement. This formulation enables the direct application of existing forward-advancing time integration schemes, and allows for the efficient incorporation of particle shifting. By applying Runge–Kutta methods, this study presents four time-marching schemes—RK1, RK2, RK3, and RK4—which are formally first- to fourth-order accurate.

Numerical investigations are conducted using five test cases: rigid body motion of a droplet, an oscillating droplet under a central force field, a Taylor–Green vortex, the merging of two co-rotating vortices, and the collision of planar opposed jets. In the first two cases, various time-marching schemes are compared, and their fundamental characteristics—including convergence and conservation properties—are investigated by observing temporal error behavior. The latter three test cases represent more complex and practical problems, where the proposed schemes are compared with the conventional Split-type scheme and FVM to evaluate their temporal accuracy, computational efficiency, and numerical stability. The results reveal that the proposed RK2, RK3, and RK4 schemes achieve second-, third-, and fourth-order temporal convergence, respectively, and exhibit substantially higher accuracy than conventional first-order schemes, leading to improved volume and energy conservation. In addition, the proposed schemes demonstrate high computational efficiency and improved numerical stability, indicating their practical value for the numerical analysis of incompressible flow.

In summary, this study has demonstrated that the proposed high-order time-marching schemes can be effectively applied to incompressible flow simulations using particle methods. While this work has focused on a specific set of time integration schemes, the same framework can be extended to construct and evaluate a broader class of methods. These findings are expected to contribute to the further development of accurate and robust computational techniques in particle-based fluid dynamics.

CRediT authorship contribution statement

Takuya Matsunaga: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests. Takuya Matsunaga reports financial support was provided by Japan Society for the Promotion of Science. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] J.J. Monaghan, Smoothed particle hydrodynamics, *Annu. Rev. Astron. Astrophys.* 30 (1992) 543–574.
- [2] S. Koshizuka, Y. Oka, Moving-particle semi-implicit method for fragmentation of incompressible fluid, *Nucl. Sci. Eng.* 123 (1996) 421–434.
- [3] T. Tamai, S. Koshizuka, Least squares moving particle semi-implicit method, *Comput. Part. Mech.* 1 (2014) 277–305.
- [4] T. Matsunaga, S. Koshizuka, Stabilized LSMPs method for complex free-surface flow simulation, *Comput. Methods Appl. Mech. Eng.* 389 (2022) 114416.
- [5] Z. Wang, T. Matsumoto, G. Duan, T. Matsunaga, Compact moving particle semi-implicit method for incompressible free-surface flow, *Comput. Methods Appl. Mech. Eng.* 414 (2023) 116168.
- [6] M. Asai, S. Fujioka, Y. Saeki, D.S. Morikawa, K. Tsuji, A class of second-derivatives in the smoothed particle hydrodynamics with 2nd-order accuracy and its application to incompressible flow simulations, *Comput. Methods Appl. Mech. Eng.* 415 (2023) 116203.
- [7] K. Shobuzako, S. Yoshida, Y. Kawada, R. Nakashima, S. Fujioka, M. Asai, A generalized smoothed particle hydrodynamics method based on the moving least squares method and its discretization error estimation, *Results Appl. Math.* 26 (2025) 100594.
- [8] S.J. Cummins, M. Rudman, A. Plx, SPH projection method, *J. Comput. Phys.* 152 (1999) 584–607.
- [9] R. Xu, P. Stansby, D. Laurence, Accuracy and stability in incompressible SPH (ISPH) based on the projection method and a new approach, *J. Comput. Phys.* 228 (2009) 6703–6725.
- [10] E.-S. Lee, C. Moulineau, R. Xu, D. Violeau, D. Laurence, P. Stansby, Comparisons of weakly compressible and truly incompressible algorithms for the SPH mesh free particle method, *J. Comput. Phys.* 227 (2008) 8417–8436.
- [11] T. Matsunaga, S. Koshizuka, Improvement of the time marching method in a particle method, *Trans. JSME* 87 (2021) 20–437. In Japanese.
- [12] S.M. Hosseini, J.J. Feng, Pressure boundary conditions for computing incompressible flows with SPH, *J. Comput. Phys.* 230 (2011) 7473–7487.
- [13] N. Trask, M. Maxey, K. Kim, M. Perego, M.L. Parks, K. Yang, J. Xu, A scalable consistent second-order SPH solver for unsteady low Reynolds number flows, *Comput. Methods Appl. Mech. Eng.* 289 (2015) 155–178.
- [14] M. Antuono, A. Colagrossi, S. Marrone, D. Molteni, Free-surface flows solved by means of SPH schemes with numerical diffusive terms, *Comput. Phys. Commun.* 181 (2010) 532–549.
- [15] M. Antuono, A. Colagrossi, S. Marrone, Numerical diffusive terms in weakly-compressible SPH schemes, *Comput. Phys. Commun.* 183 (2012) 2570–2580.
- [16] P.N. Sun, A. Colagrossi, S. Marrone, A.M. Zhang, The δ plus-SPH model: simple procedures for a further improvement of the SPH scheme, *Comput. Methods Appl. Mech. Eng.* 315 (2017) 25–49.
- [17] H.G. Lyu, P.N. Sun, X.T. Huang, S.H. Chen, A.M. Zhang, On removing the numerical instability induced by negative pressures in SPH simulations of typical fluid-structure interaction problems in ocean engineering, *Appl. Ocean Res.* 117 (2021) 102938.
- [18] F. Capuano, G. Coppola, L. Rández, L.D. Luca, Explicit Runge-Kutta schemes for incompressible flow with improved energy-conservation properties, *J. Comput. Phys.* 328 (2017) 86–94.
- [19] M. Karam, J.C. Sutherland, T. Saad, Low-cost Runge-Kutta integrators for incompressible flow simulations, *J. Comput. Phys.* 443 (2021) 110518.
- [20] A.B. Aithal, M. Tipirneni, A. Ferrante, Temporal accuracy of FastRK3, *J. Comput. Phys.* 475 (2023) 111853.
- [21] M. Karam, T. Saad, On the theory of fast projection methods for high-order Navier-Stokes solvers, *J. Comput. Phys.* 495 (2023) 112557.
- [22] T. Matsunaga, A. Södersten, S. Koshizuka, T. Hosaka, E. Ishii, Axisymmetric free-surface flow simulation using the moving surface mesh particle method and application to drop formation, *J. Comput. Phys.* 463 (2022) 111298.
- [23] S. Marrone, A. Colagrossi, D.L. Touzé, G. Graziani, Fast free-surface detection and level-set function definition in SPH solvers, *J. Comput. Phys.* 229 (2010) 3652–3663.
- [24] N. Trask, M. Maxey, X. Hu, Compact moving least squares: an optimization framework for generating high-order compact meshless discretizations, *J. Comput. Phys.* 326 (2016) 596–611.
- [25] D. Mirzaei, R. Schaback, M. Dehghan, On generalized moving least squares and diffuse derivatives, *IMA J. Numer. Anal.* 32 (2012) 983–1000.
- [26] J.C. Butcher, A history of Runge-Kutta methods, *Appl. Numer. Math.* 20 (1996) 247–260.
- [27] J.C. Butcher, *Numerical Methods for Ordinary Differential Equations*, John Wiley & Sons, 2016.
- [28] J.L. Guermond, P. Mineev, J. Shen, An overview of projection methods for incompressible flows, *Comput. Methods Appl. Mech. Eng.* 195 (2006) 6011–6045.
- [29] J.J. Monaghan, A. Rafee, A simple SPH algorithm for multi-fluid flow with high density ratios, *Int. J. Numer. Methods Fluids* 71 (2013) 537–561.
- [30] A. Khayyer, H. Gotoh, Y. Shimizu, Comparative study on accuracy and conservation properties of two particle regularization schemes and proposal of an optimized particle shifting scheme in ISPH context, *J. Comput. Phys.* 332 (2017) 236–256.
- [31] P.P. Wang, Z.F. Meng, A.M. Zhang, F.R. Ming, P.N. Sun, Improved particle shifting technology and optimized free-surface detection method for free-surface flows in smoothed particle hydrodynamics, *Comput. Methods Appl. Mech. Eng.* 357 (2019) 112580.
- [32] P. Meunier, U. Ehrenstein, T. Lewke, M. Rossi, A merging criterion for two-dimensional co-rotating vortices, *Phys. Fluids* 14 (2002) 2757–2766.
- [33] C. Cerretelli, C.H.K. Williamson, The physical mechanism for vortex merging, *J. Fluid Mech.* 475 (2003) 41–77.
- [34] P. Meunier, S.L. Dizès, T. Lewke, Physics of vortex merging 6 (2005) 431–450. *Comptes Rendus Phys.*
- [35] C. Josserand, M. Rossi, The merging of two co-rotating vortices: a numerical study, *Eur. J. Mech. B/Fluids* 26 (2007) 779–794.
- [36] G. Taylor, Oblique impact of a jet on a plane surface, *Philosophical Trans. Royal Soc. London A: Math. Phys. Sci.* 260 (1966) 96–100.

- [37] L.M. Milne-Thomson, *Theoretical Hydrodynamics*, Macmillan, 5th edition edition, 1968.
- [38] D. Molteni, A. Colagrossi, A simple procedure to improve the pressure evaluation in hydrodynamic context using the SPH, *Comput. Phys. Commun.* 180 (2009) 861–872.
- [39] A. Khayyer, H. Gotoh, Enhancement of stability and accuracy of the moving particle semi-implicit method, *J. Comput. Phys.* 230 (2011) 3093–3118.